Hi, my name is Michelle

I bridge the gap between design and implementation.

I am currently studying for a Bachelor's degree in **Computer Science, specializing in Design and Management**, at the University of Applied Sciences and Arts Northwestern Switzerland (FHNW).

I focus on **UI/UX** through the lens of technical feasibility. As I understand the code behind the components, I design interfaces that are as easy to build as they are to use.

and this is my

Portfolio

Contact me

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LinkedIn

michelleree

🎓 Education

09.2023 - now

BSc Computer Science - iCompetence, Full-time Specialization in Design and Management

University of Applied Sciences and Arts Northwestern Switzerland (FHNW), Windisch

- Development of prototypes with Figma
- Implementation of interactive websites with HTML, CSS, and JavaScript in modules and interdisciplinary projects
- Bachelor's project: Development of acceptance strategies for the use of care robots in retirement homes

08.2022 - 06.2023

Technical Vocational School (Berufsmatura)

Berufsschule Aarau

- Developing and designing an app using a web-based platform (no direct programming)

08.2019 - 06.2022

Vocational School

Allgemeine Gewerbeschule, Basel

💼 Experience

08.2019 - 06.2022

Chemical Laboratory Assistant EFZ

F. Hoffmann-La Roche, Basel/Kaiseraugst

- Evaluation and validation of measurement data with Empower (data analysis software)
- Close collaboration with teams in the laboratory
- Active contribution of suggestions for improvement in team meetings

🛠️ Skills

🔧 Tools

Figma

Adobe Illustrator

Adobe After Effects

💻 Development

HTML

CSS

JavaScript

🎨 Design

UX Design

UI Design

Prototyping

User Research

🤝 Soft Skills

Adaptability

Organization

Teamwork

🌐 Languages

German

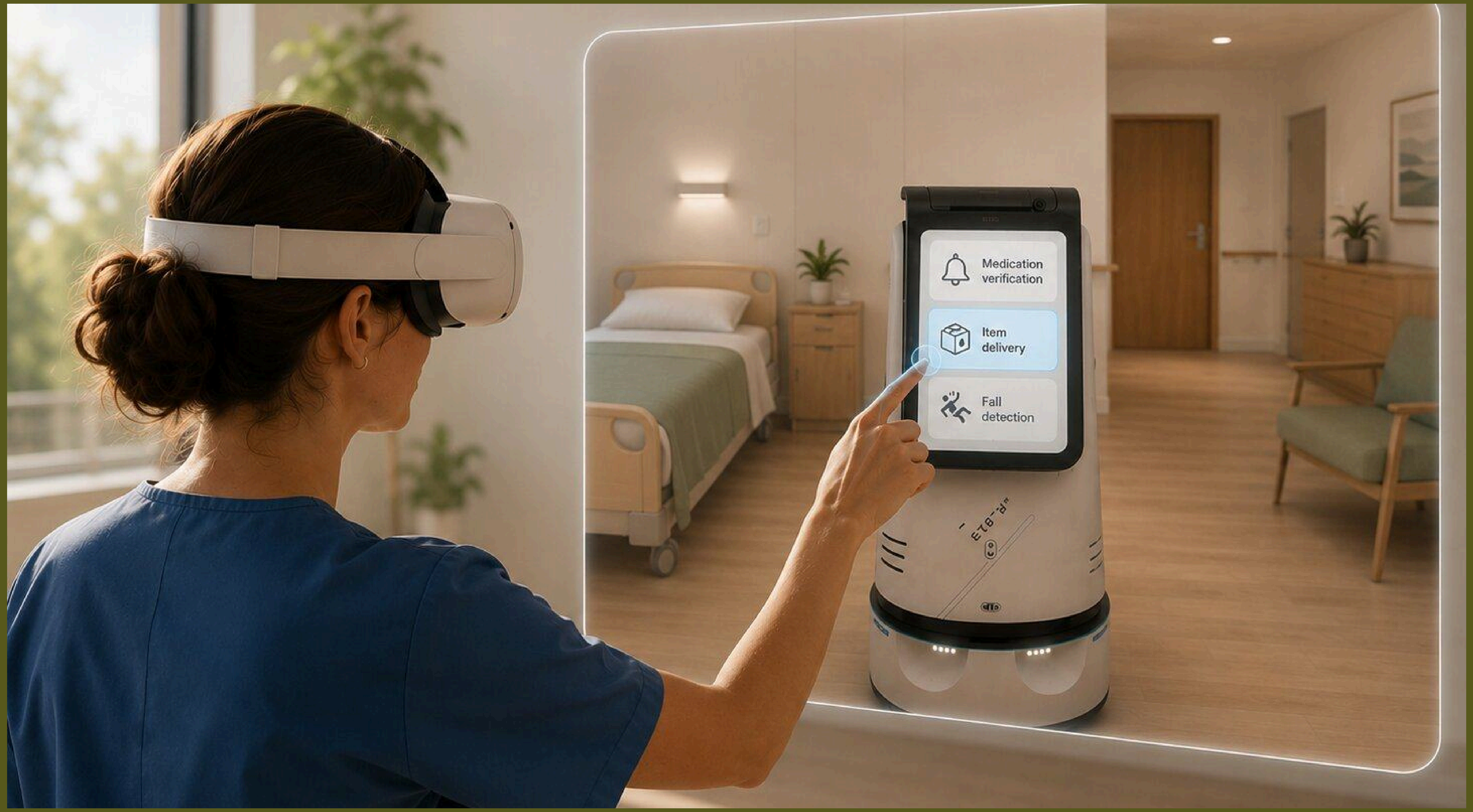
English

Spanish

French

🤖 Exploring Acceptability and Integration Strategies for Care Robots in Senior Homes

Designing responsible human-robot collaboration for senior care through research, VR prototyping, and user testing.



Module:	Bachelor’s thesis / IP6 (Bachelor’s Project Semester 6)
Team:	2 people (pair programming + individual work)
Duration:	1 semester (approx. 360h per person)
Role:	UX/UI, Interaction Design, VR Prototyping & Evaluation
Tools:	Unity, C#, Meta Quest Pro, Miro, Figma, ChatGPT (interview & evaluation analysis, brainstorming ideas)
Client:	Spitex Unicare AG

Detailed description of the project:
🔗 Project Care Robots in Senior Homes

🧐 Problem

Healthcare systems are facing increasing challenges due to an ageing population and a growing demand for long-term care. Caregivers face heavy workloads, physically demanding tasks, and less time to spend with patients. Robots could take over repetitive routine work, but having the technology is not the same as having people accept it. Whether caregivers are willing to work with a robot ultimately comes down to trust and transparency.

🎯 Goal

The goal of this project was to investigate how care robots can be introduced into senior homes in a meaningful and responsible way while supporting caregivers rather than replacing them.

👩🏻‍💻 My Contribution

- Conducted user research and analysis through interviews with professional caregivers at Spitex Unicare
- Synthesized findings into personas, user needs, and pain points
- Defined a structured task evaluation and a 3-phase integration framework
- Designed a VR prototype for three representative care scenarios using Unity and Meta Quest Pro
- Planned and conducted usability testing with 4 participants, collecting quantitative and qualitative data

🔎 Research Insights

Robots in Healthcare - General Findings:

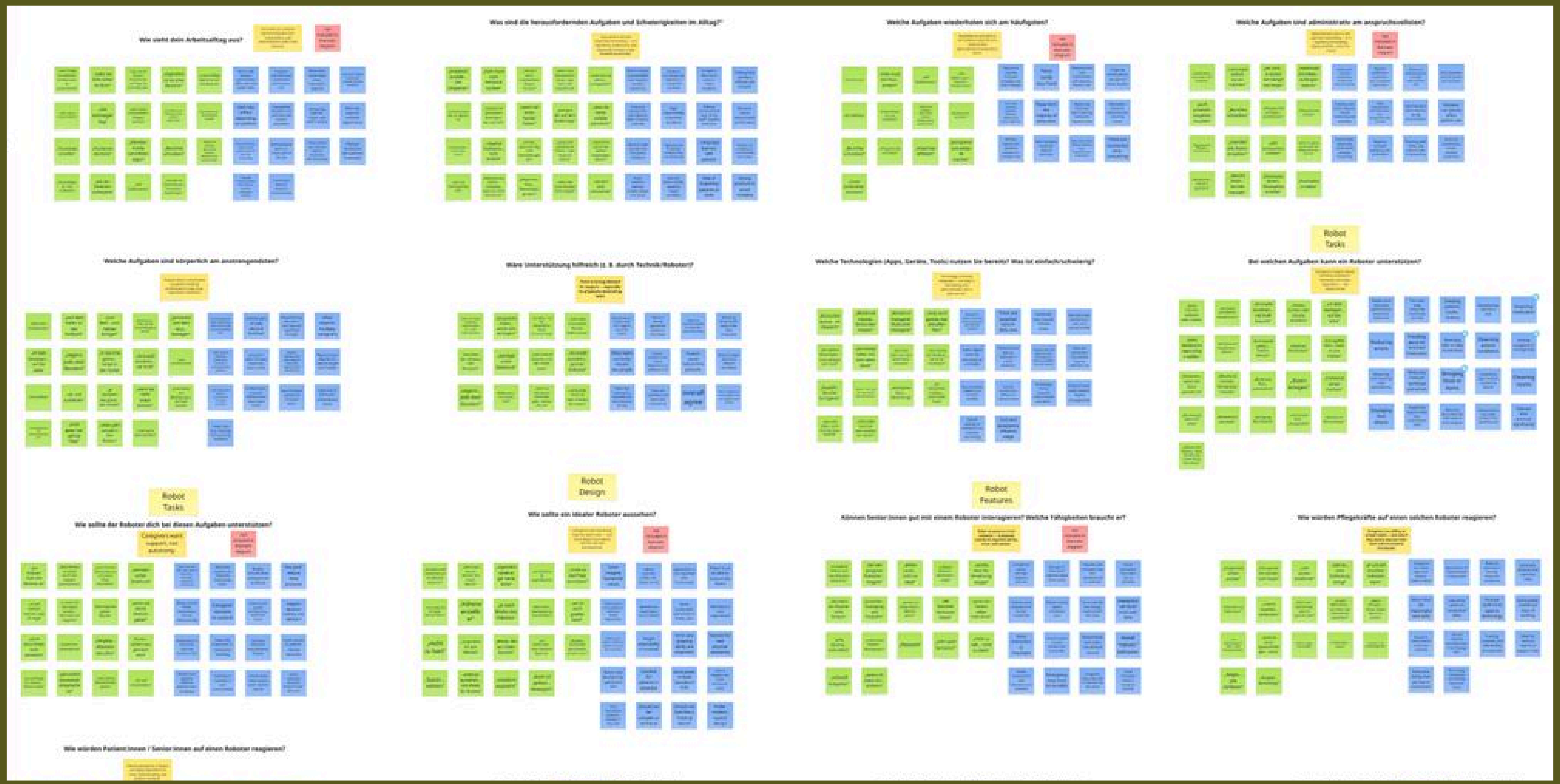
- Existing robots like Lio, Moxi, and Care-O-Bot 4 demonstrate capabilities in logistics, monitoring, and social interaction.
- **Common strengths:** repetitive tasks, object delivery, multimodal interaction.
- **Common limitations:** complex physical tasks, autonomous decision-making, reliability concerns.
- Successful integration depends on gradual introduction, transparency, and maintaining human control.

Caregiver Interviews (Spitex Unicare):

- Caregivers are open to robotic support for repetitive and physically demanding tasks but want to remain in control of all clinical decisions.
- Physical burden (mobilising patients, putting on compression socks) and time pressure are major challenges.
- Trust is fragile because errors or a lack of transparency can quickly reduce acceptance.
- **Tasks should be introduced gradually:** first simple logistical support, then clinical verification, finally safety-critical monitoring.
- Medication errors are a real concern—robots could help verify but not administer medication.
- Patients with dementia could become confused or frightened by robots.
- **Design preferences:** compact, height-adjustable, simple interface with buttons and display.

🔄 Design Process

User Research (Interviews & Literature) → Personas → Brainstorming → Requirements → Task Evaluation & Selection → Storyboards & User Journeys → Integration Framework → Prototyping in VR → Usability Testing



Excerpt From the Interview Analysis

One of Two Personas

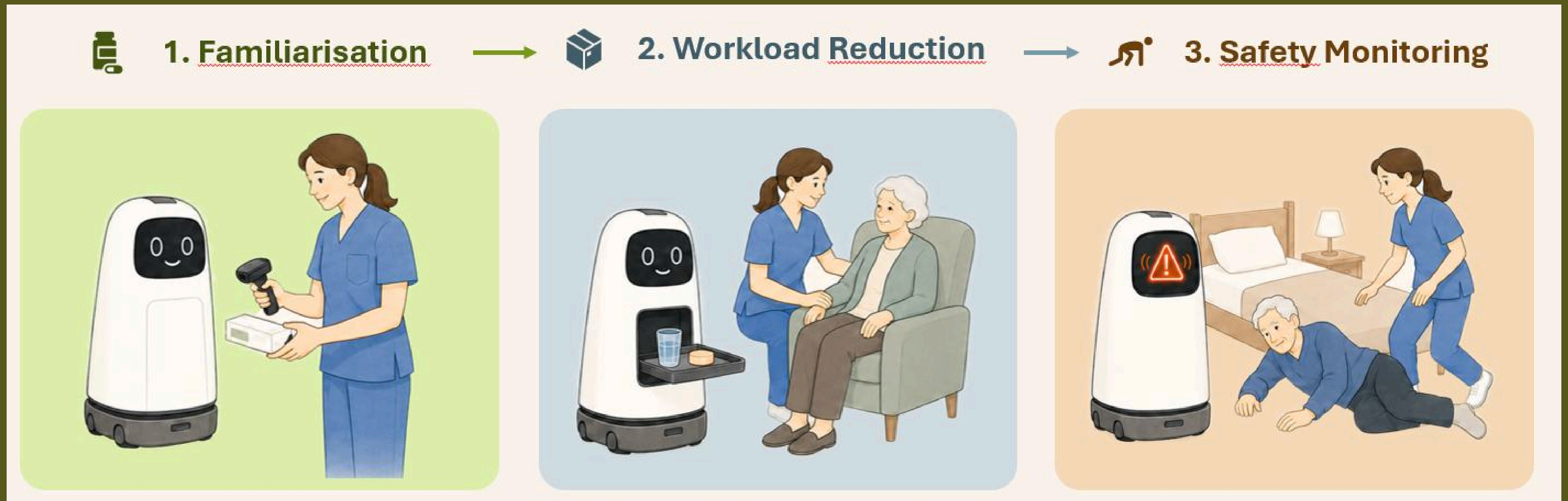
Story Board Scenario 1

Journey phases	start shift	monitor patients	task overload	reminder triggered	action taken	task completed	documentation
Jobs to be done	understand daily schedule	check patient needs	prioritize multiple tasks	receive accurate medication reminder with latest prescription information	prepare medication	administer medication	record completion
Context & Touchpoints	nurse station, schedule system	patient room	multiple patients, interruptions	robot / mobile notification	medication cart	patient bedside	digital system / documentation tool
Gains	clear overview of tasks	efficient workflow	support prioritization	timely reminder reduces forgetting	fast preparation	safe medication delivery	accurate records
Pains	complex schedules	constant interruptions	stress / time pressure	risk of missing medication	searching for meds takes time, risk of outdated prescription, wrong dosage risk	risk of errors	documentation effort
Emotion							

User Journey Scenario 1

🚀 Key Features

- **Phased Integration Framework:** Defines a progression from familiarization (medication reminder) to workload reduction (item delivery) to safety-critical support (fall detection).
- **VR Simulation:** Interactive implementation of three care scenarios allowing safe, cost-effective evaluation of interaction concepts.
- **Medication Verification:** The robot acts as a verification tool, helping to prevent medication errors while keeping the caregiver in control.
- **Item Delivery:** The robot autonomously retrieves requested items, reducing workload and allowing more time for patient interaction.
- **Fall Detection:** The robot monitors for falls and immediately alerts the caregiver, improving patient safety without replacing human judgment.



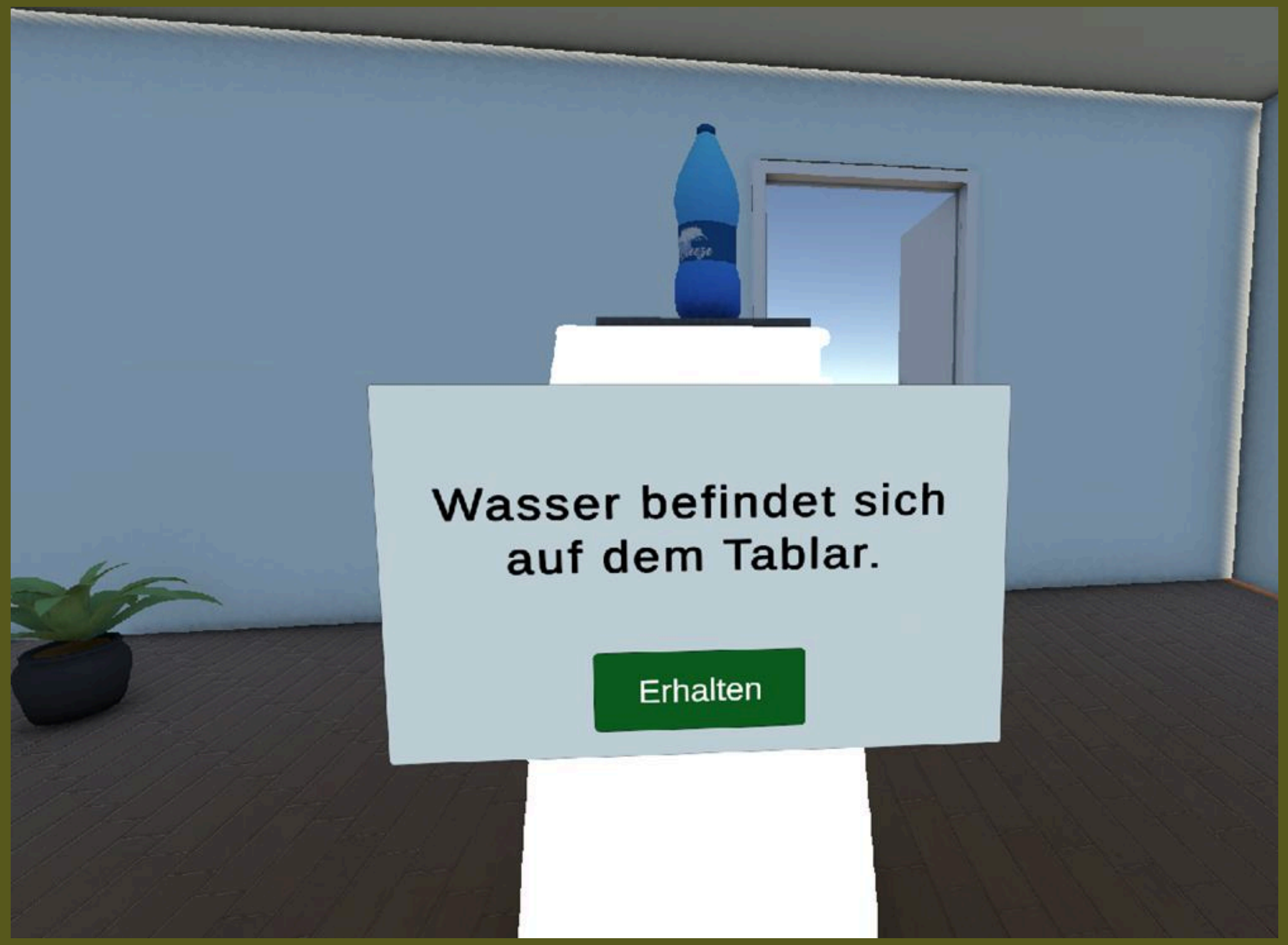
📝 User Testing Results

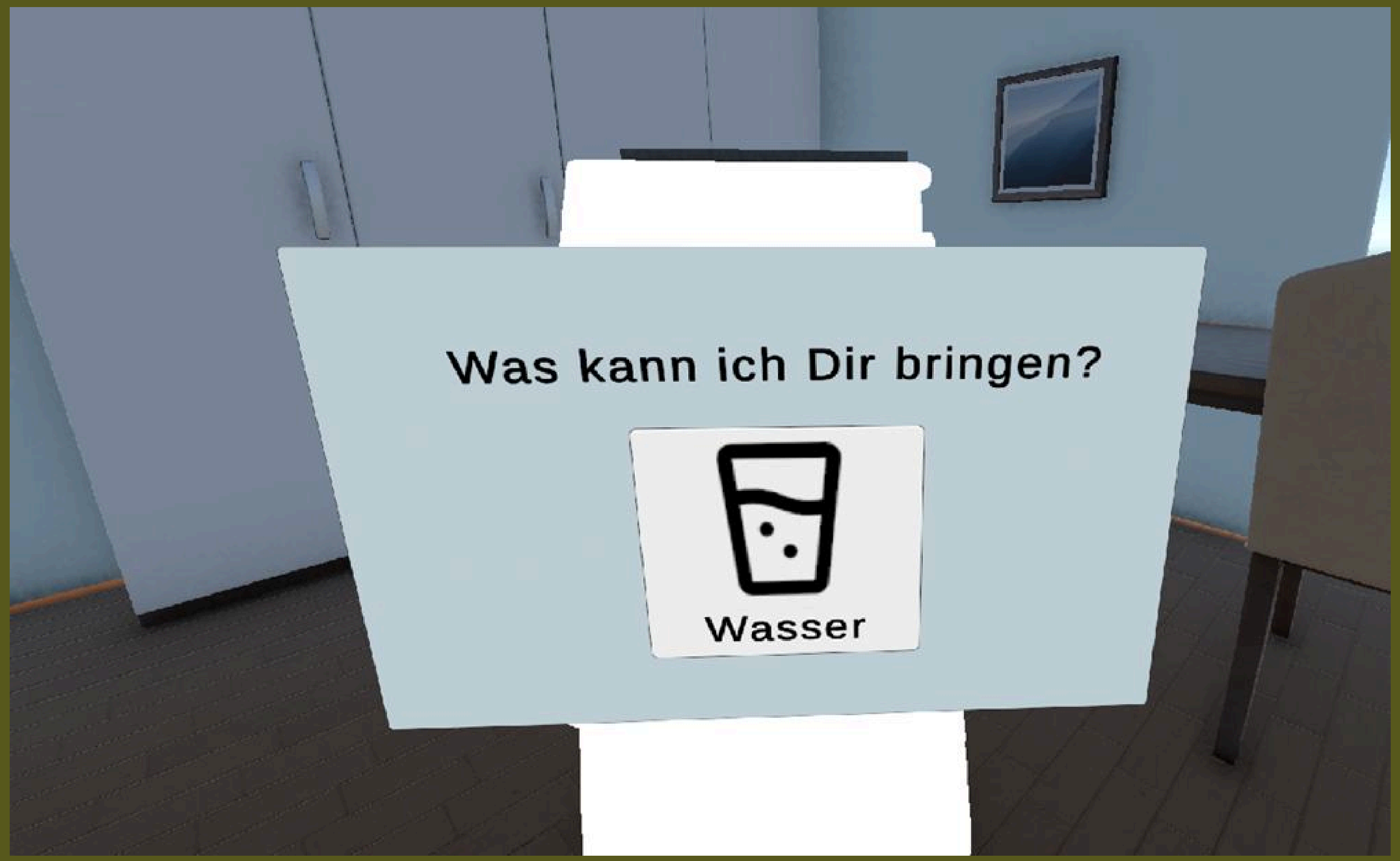
- Participants preferred a gradual introduction, starting with logistical support tasks.
- Trust was highest for clearly defined tasks with low clinical risk (item delivery).
- Trust was conditional on transparency and reliability, particularly for safety-critical functions (fall detection).
- The VR prototype was an effective tool for evaluating interaction concepts before physical implementation.

💡 Reflection

This project taught me the importance of designing for real-world acceptance, not just technical feasibility. The biggest challenge was translating complex user needs into a structured framework and an interactive prototype that effectively communicated the concept. The evaluation showed that trust is fragile and context-dependent, and it must be earned through clear, reliable, and supportive design. If I were to continue this work, I would:

- Implement voice interaction for more natural communication.
- Test the framework in a real care environment with a physical robot.
- Conduct a longitudinal study to see how trust develops over time.





🗺️ Interactive 3D Map of Relocation Process

Making city exploration intuitive through 3D interaction and light gamification

Module: IP5 (Project Semester 5)
Team: 2 people (pair programming + individual work)
Duration: 1 semester (approx. 180h per person)
Role: UX/UI, Interaction Design & Frontend Implementation
Tools: Figma, JavaScript, Three.js, MapLibre, OpenStreetMap, ChatGPT (debugging complex code, user evalution analysis)
Client: SwissPlatz

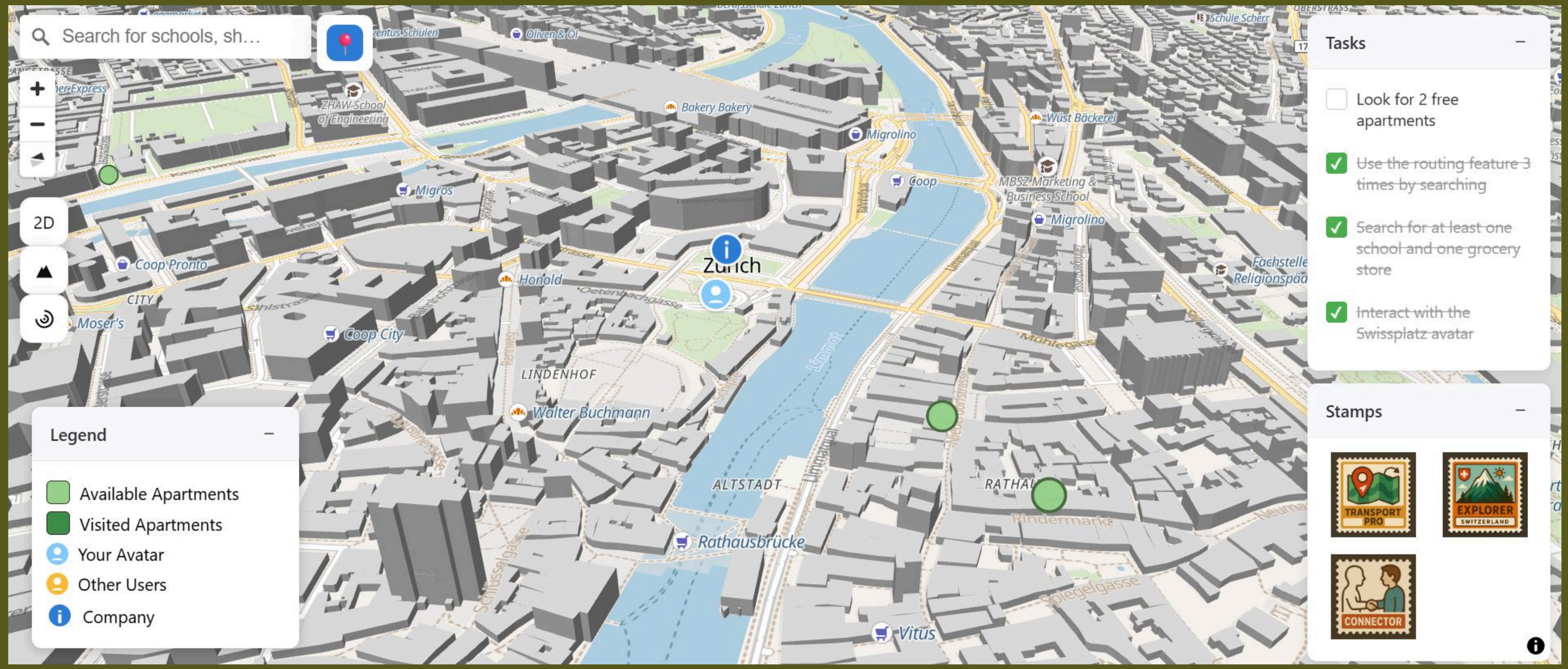
Detailed description of the project:
🔗 Project Relocation

🧐 Problem

Relocating to a new city is overwhelming. Existing platforms focus on information delivery but often lack spatial understanding and emotional support. Newcomers struggle to grasp neighborhood structures, building density, and spatial relationships.

🎯 Goal

The goal of this project was to design a 3D interactive platform that helps users explore cities intuitively, understand spatial relationships, and feel more confident about their relocation decisions.



👩🏻‍💻 My Contribution

- Conducted user research to understand relocation pain points
- Created personas, storyboards, and user flows
- Designed prototypes in Figma
- Implemented the 3D map interface using MapLibre
- Built the avatar movement system and NPC interactions
- Designed and integrated gamification elements (tasks + stamps)
- Planned and conducted usability testing

🔎 Research Insights

- Users need spatial context (building heights, neighborhood density and structure)
- Exploration should feel intuitive, not like work
- Gamification can motivate without distracting

🔄 Design Process

User research → Persona → Brainstorming → Sketches → User flows → Figma prototype → Implementation → User testing

🚀 Key Features

- **3D Map:** Extruded buildings for spatial understanding
- **Avatar System:** User representation with movement
- **Gamification:** Tasks + stamps that reward exploration
- **Information Overlays:** Apartments, NPCs, POIs

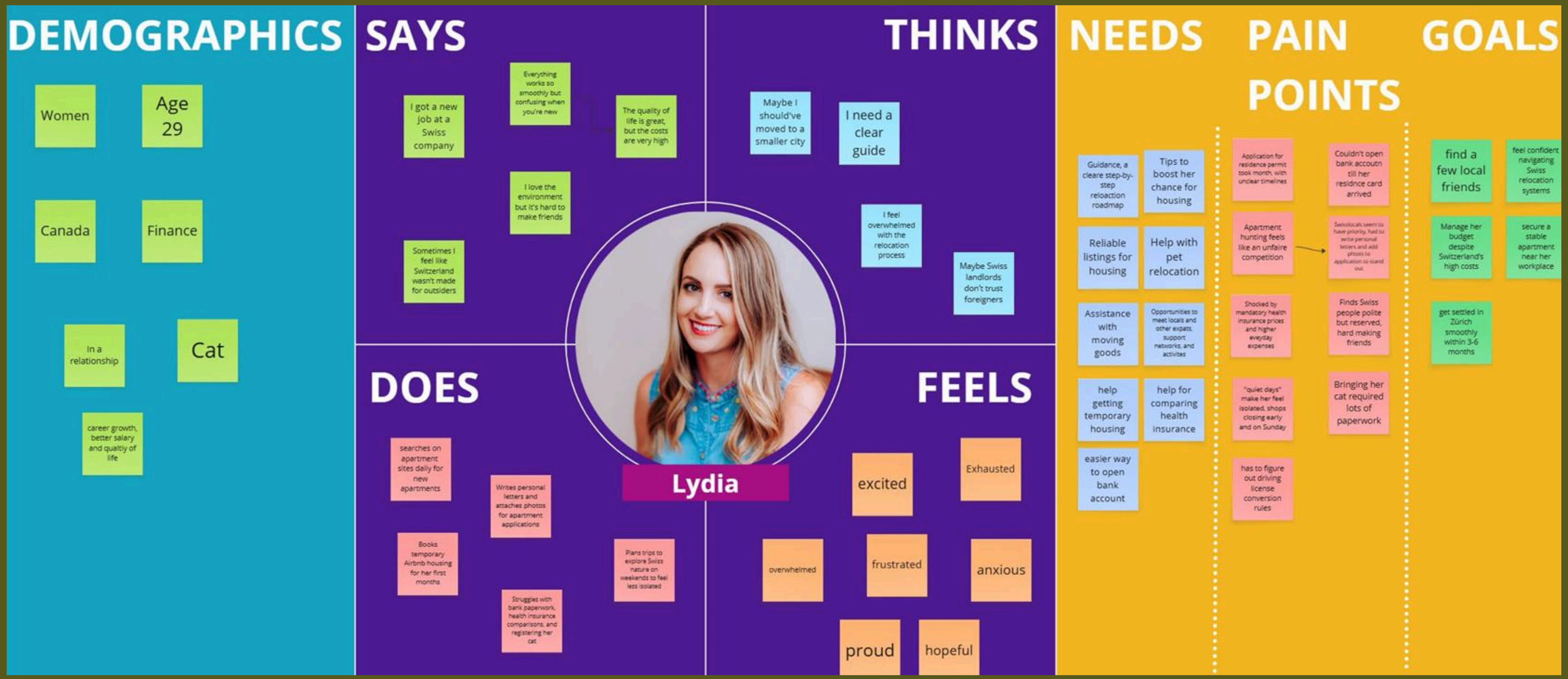
📝 User Testing Results

- 3D map improved spatial understanding
- Gamification increased engagement without distraction
- Users preferred 3D for exploration, 2D for efficiency

💡 Reflection

This project taught me how to balance innovation with usability. The 3D map genuinely helped users understand neighborhoods better, but I learned that familiar interactions (like Google Maps-style navigation) are essential for adoption. If I did it again, I would:

- Add a quick onboarding tutorial
- Include more filtering options
- Test with a more diverse user group



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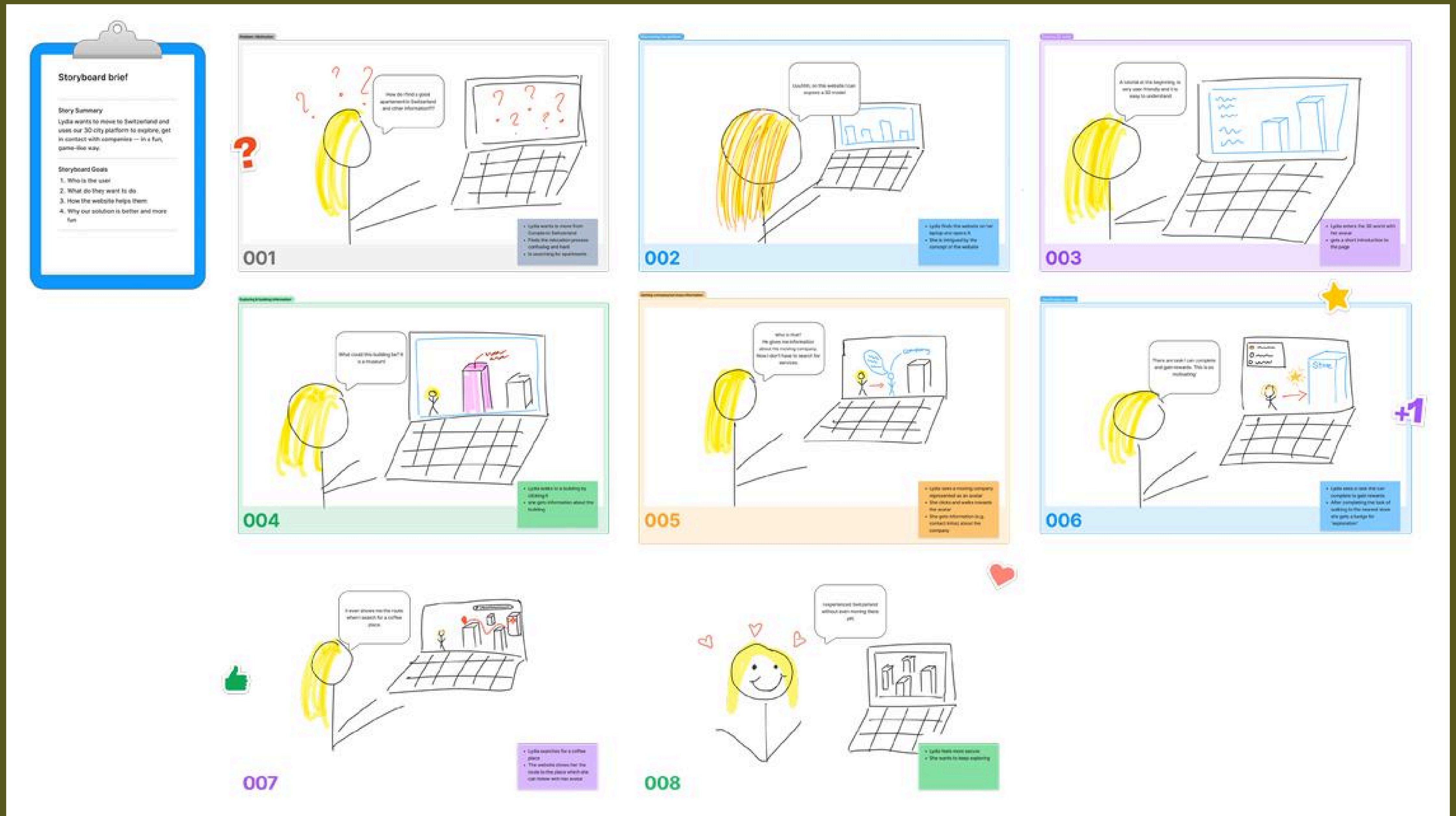
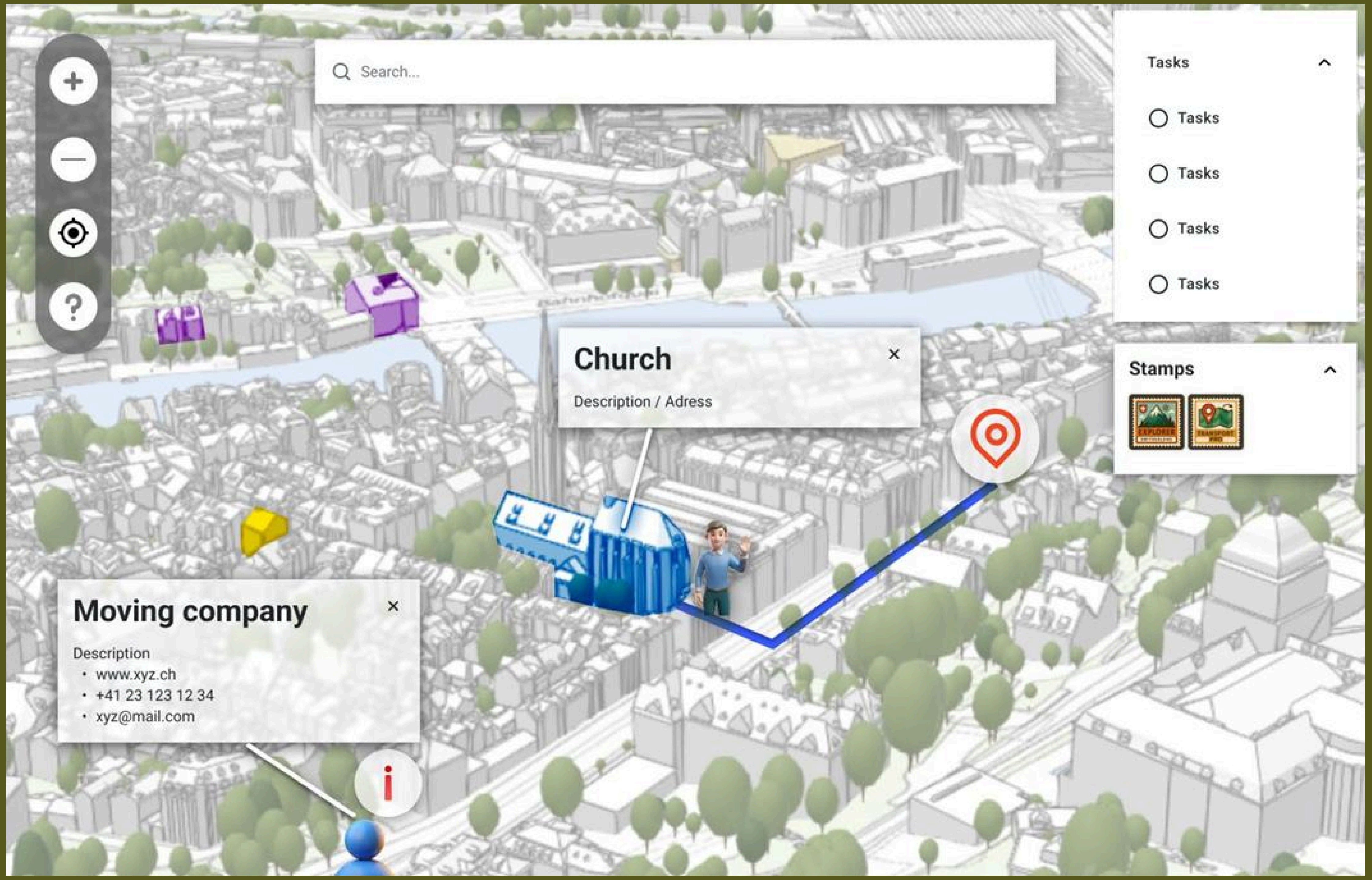
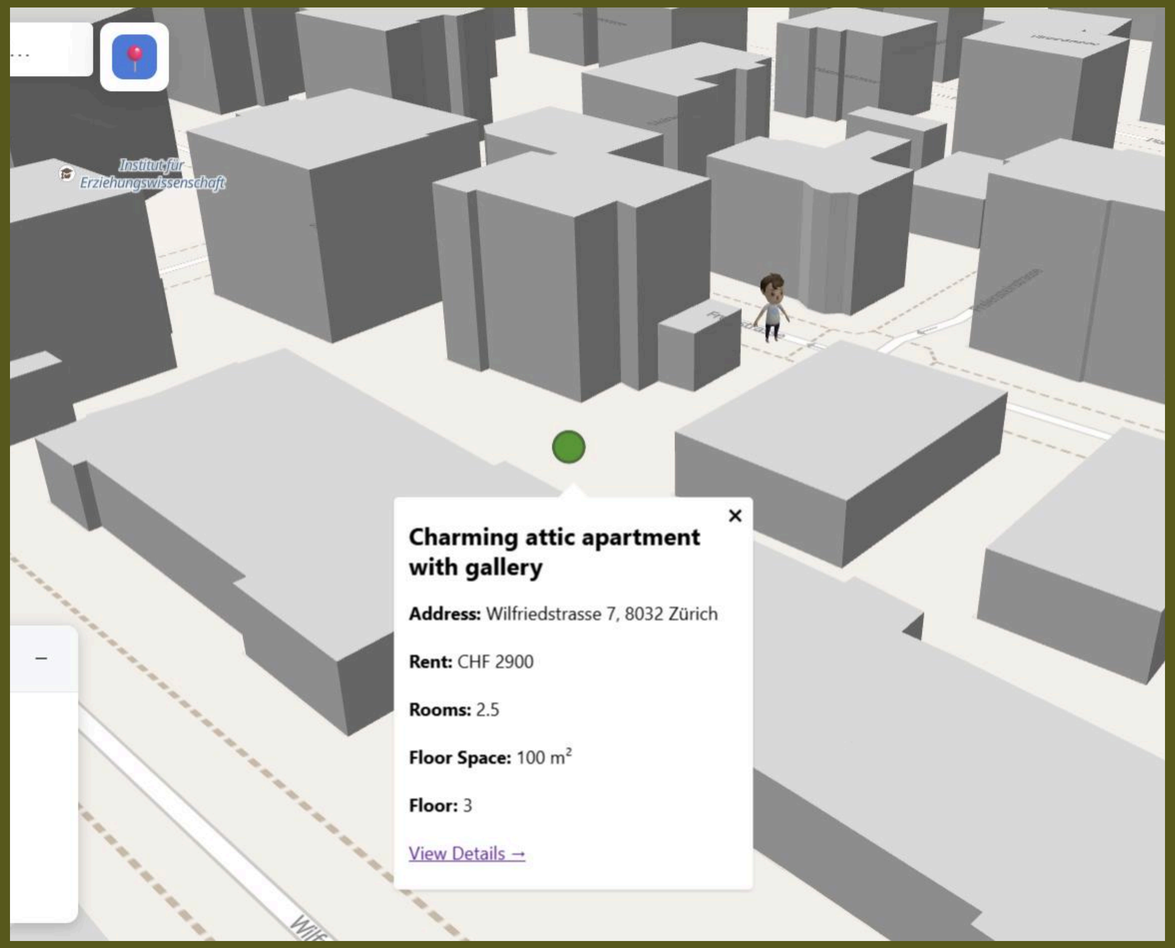
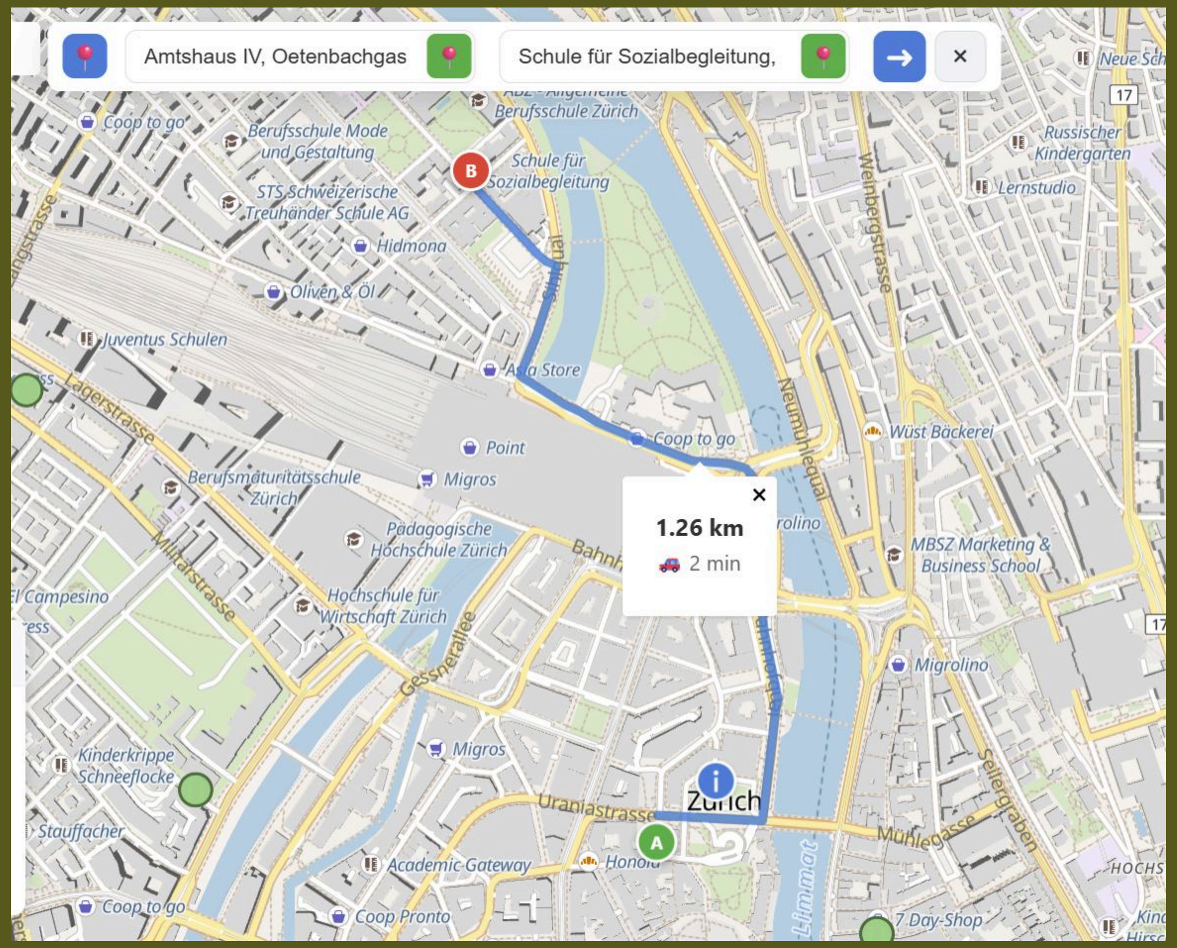


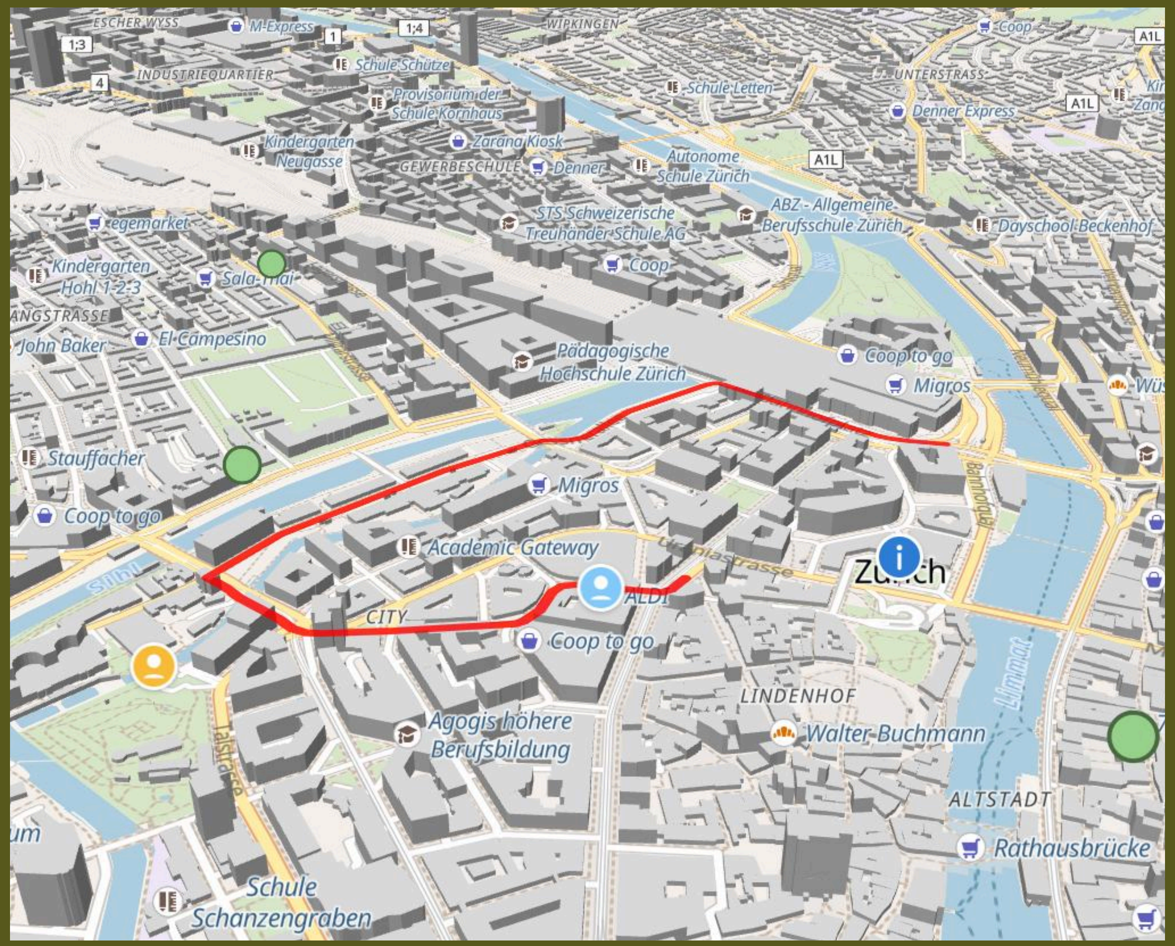
Created Story Board

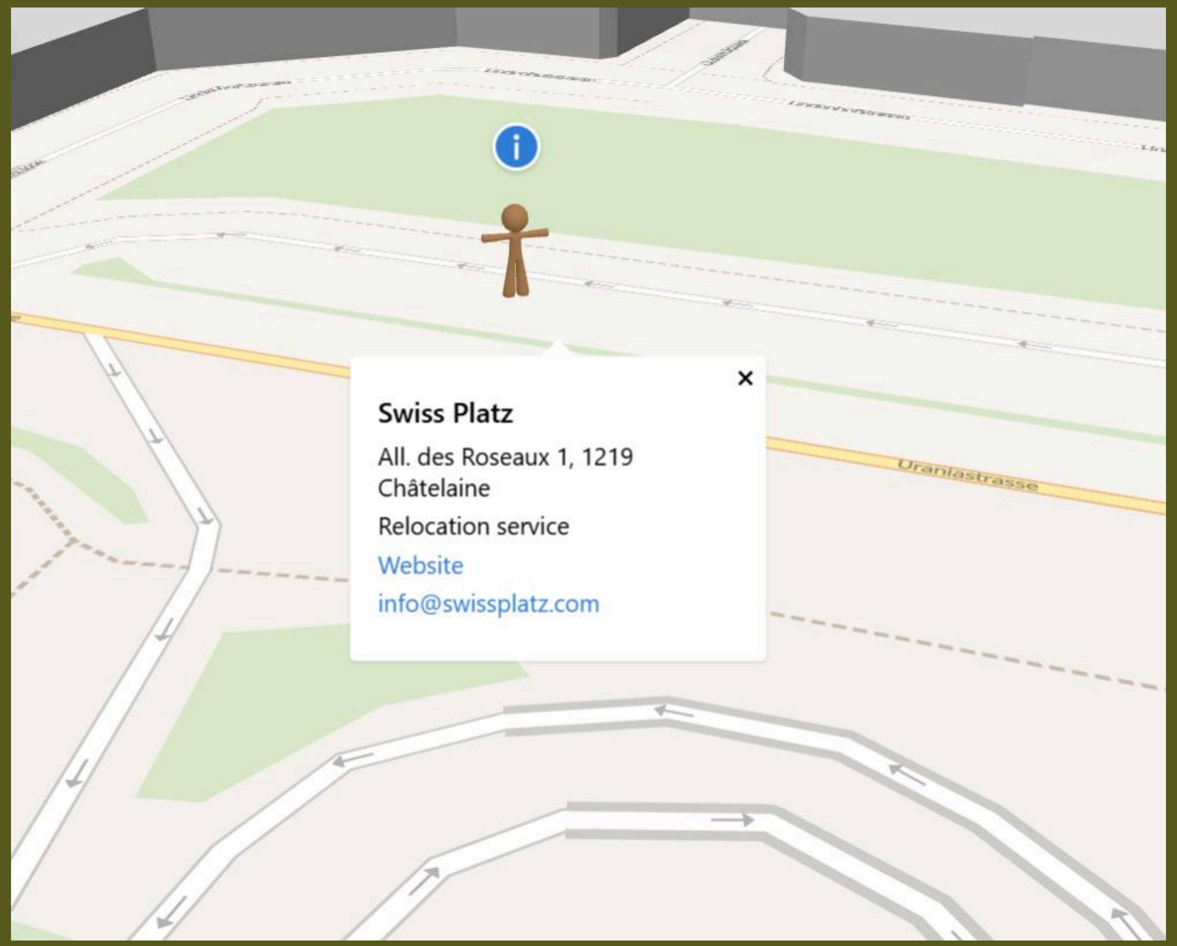


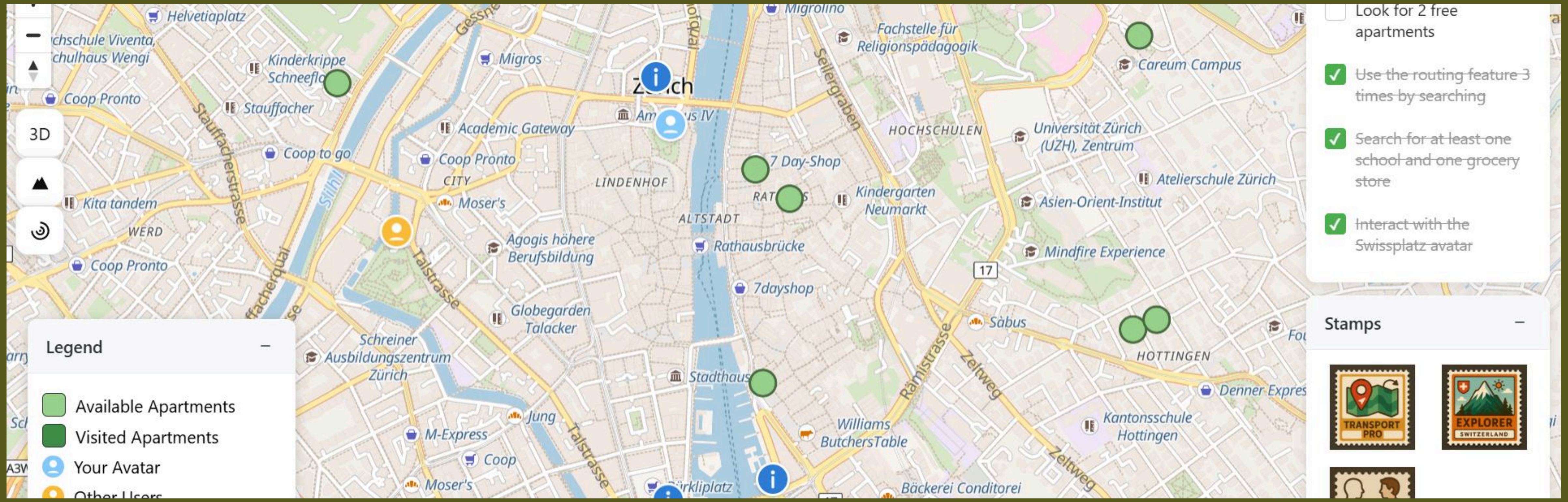
Figma Prototype











🧱 Power Tower Challenge

Making Switzerland's energy future tangible through interactive learning

Module: IP34 (Project Semester 3 & 4)
Team: 6 people
Duration: 2 semesters
Role: Product Owner, UX/UI, Frontend Developer
Tools: Figma, React, JavaScript, TypeScript, HTML, Tailwind CSS, Python, Next.js
Client: WWF

Detailed description of the project:
🔗 Project Power Tower

🧐 Problem

WWF created a physical game where players build a tower representing Switzerland's 2035 energy needs. But the physical game lacked guidance and feedback because the players couldn't understand the impact of their choices without expert help.

🎯 Goal

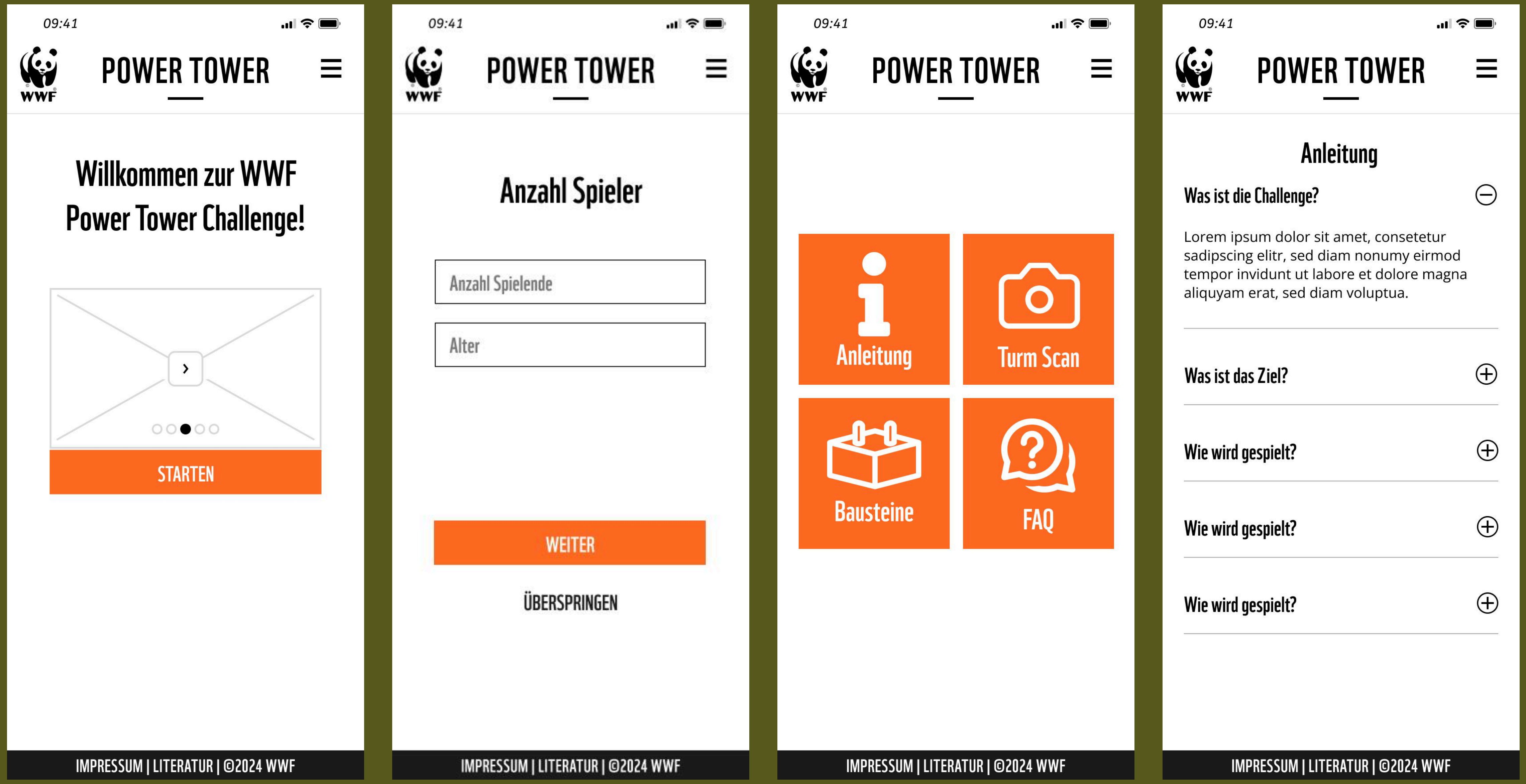
The goal of this project was to design a web app that lets users scan their physical tower, get instant feedback on sustainability, and learn about energy choices independently.

👩🏻‍💻 My Contribution

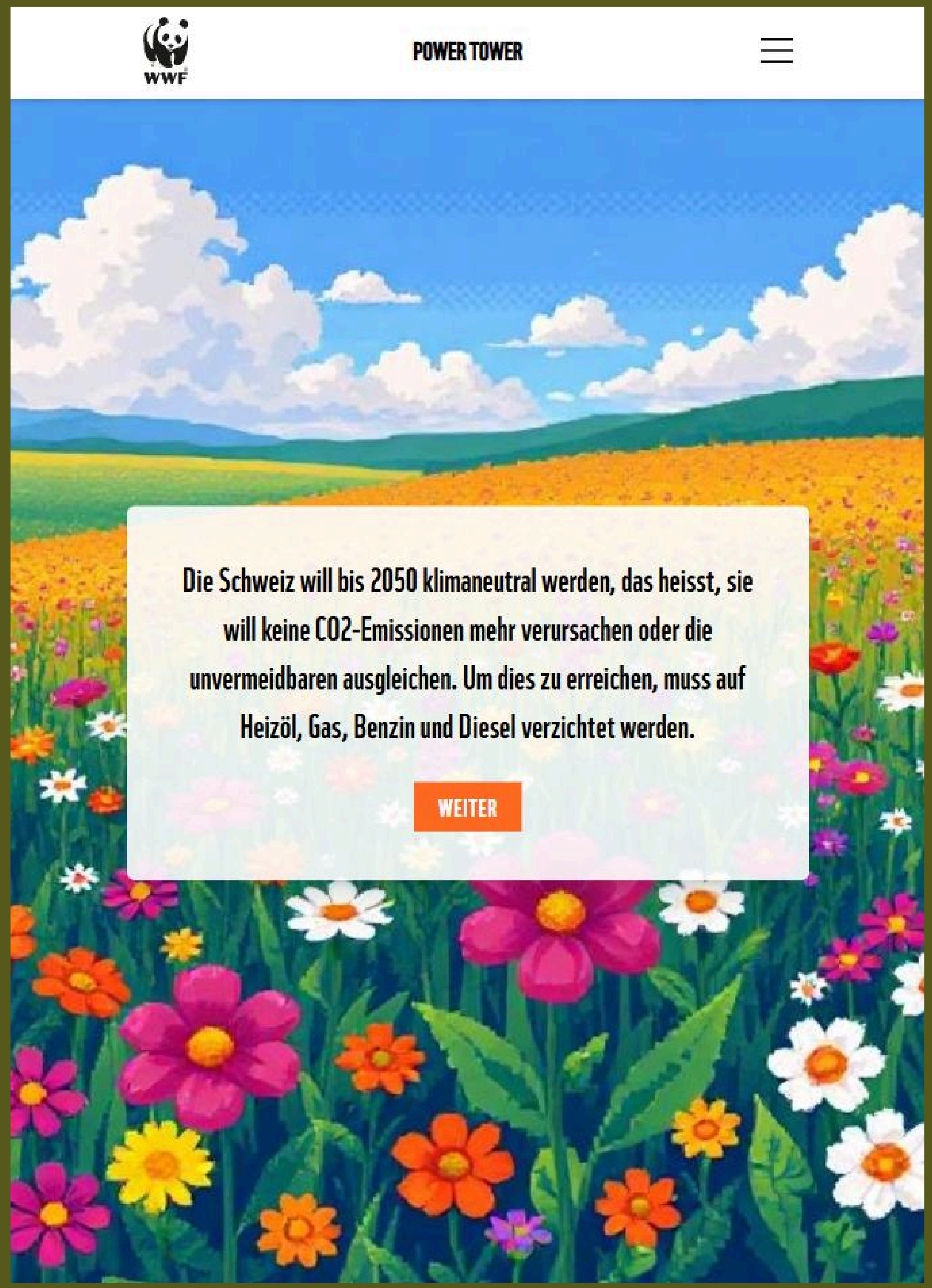
- Led UI/UX design from concept to high-fidelity prototypes
- Designed the scanning flow and feedback interface
- Contributed to frontend implementation
- Collaborated with client on feature prioritization

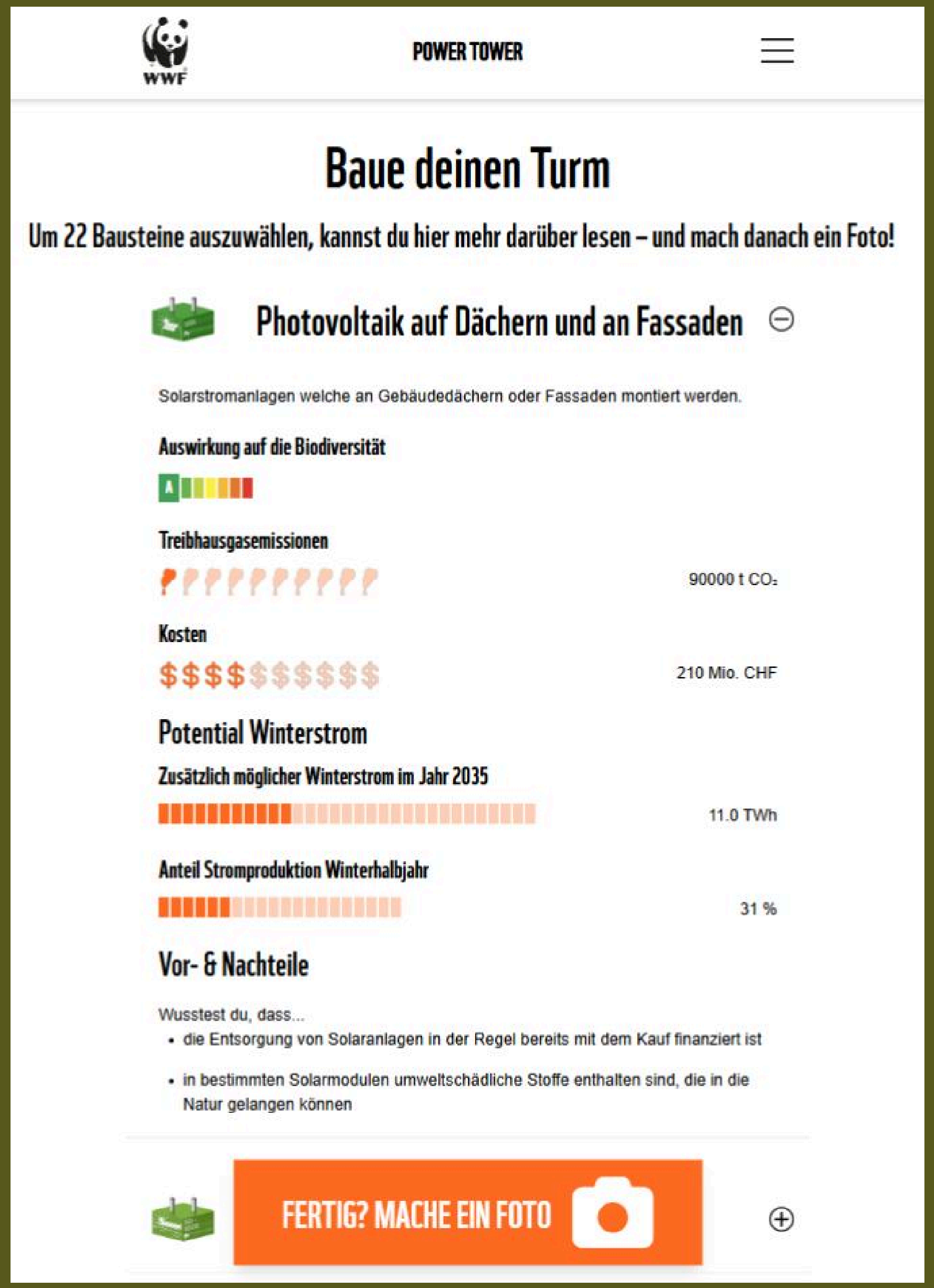
🔄 Design Process

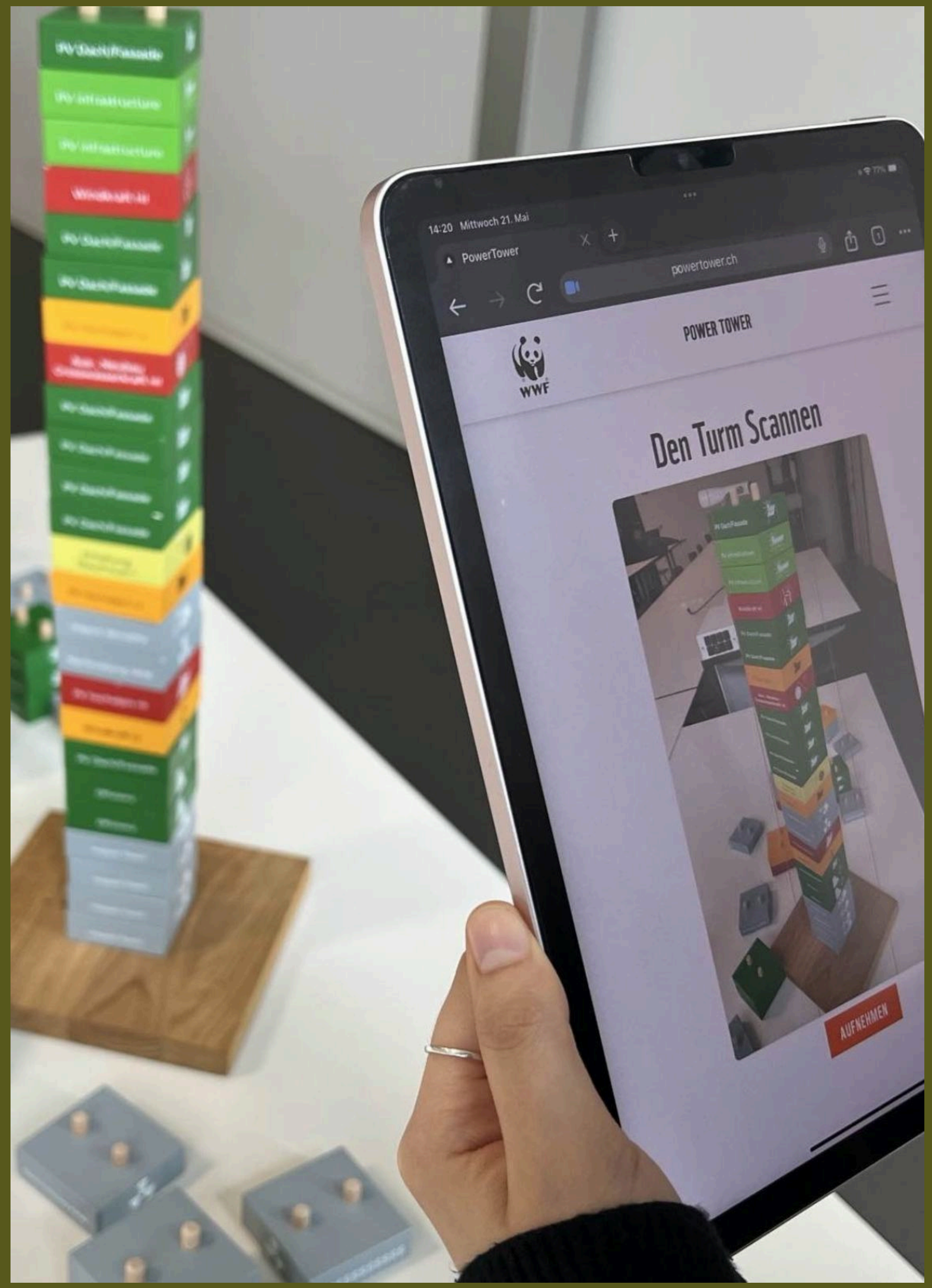
User stories → Figma prototype → Implementation → User testing → Iteration

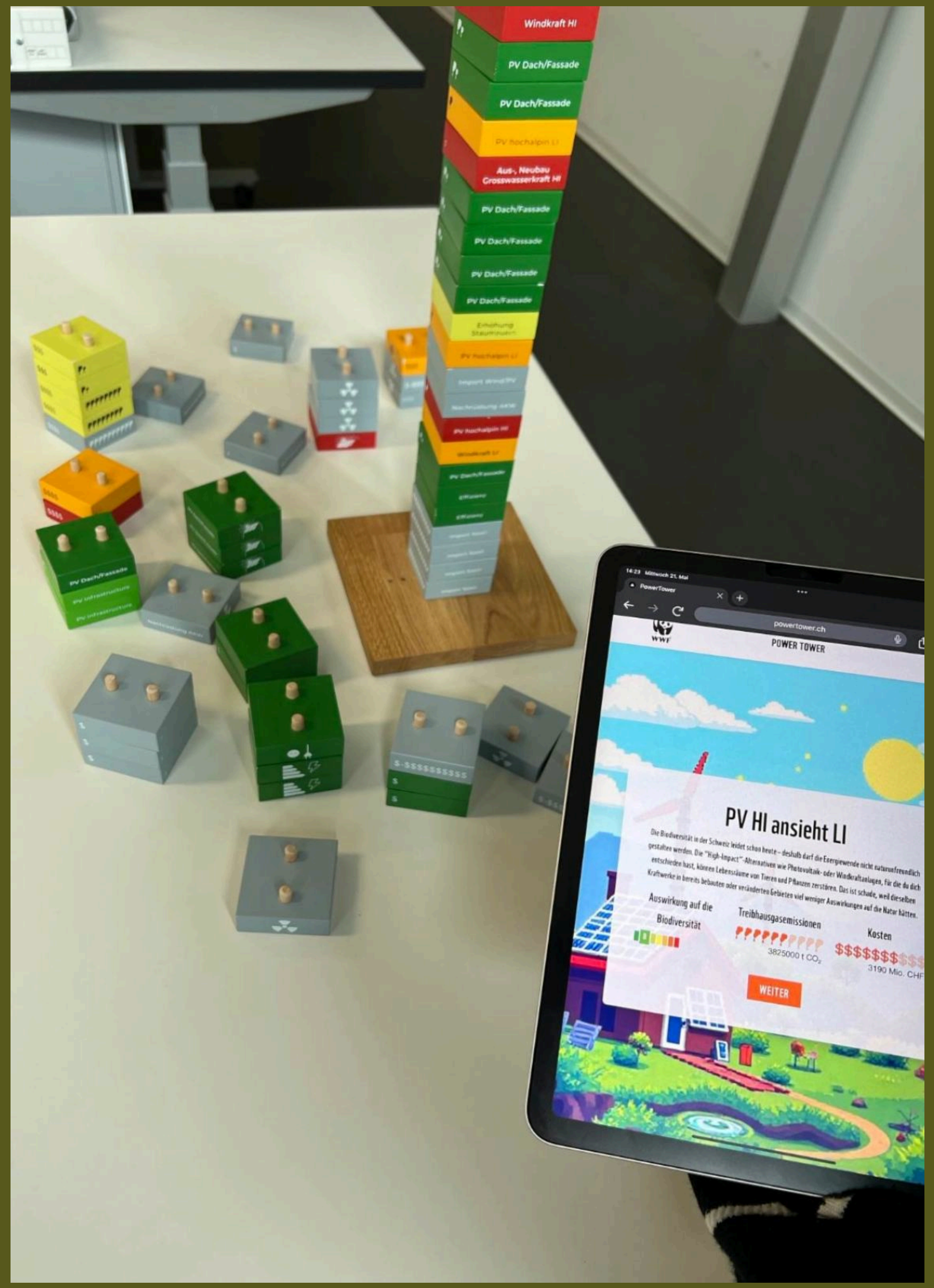


Protoype Frames on Figma.









🚀 Key Features

- **Introduction:** Background behind the game
- **Building Blocks Overview:** Details of all available building blocks
- **Scanning Feature:** Scanning physical tower with device camera
- **Tower Evaluation:** Feedback on physical tower
- **Quiz:** Testing the knowledge

💡 Reflection

This project taught me how to bridge physical and digital experiences. The biggest challenge was designing a feedback system that was both accurate and understandable. Working with WWF also showed me the value of client collaboration and being proactive with ideas.

⚡ Powerhouse

An interactive dollhouse that teaches energy efficiency through playful experimentation

Module: IP12 (Project Semester 1 & 2)
Team: 7 people
Duration: 2 semesters
Role: UI/UX, Interaction, Project Manager
Tools: Java, JavaFX, CSS, Pi4J, Raspberry Pi
Client: FHNW, Primeo

Detailed description of the project:
🔗 Project Powerhouse

🧐 Problem

Young people struggle to connect abstract energy concepts to their daily lives. How can we make energy efficiency tangible and engaging for teenagers visiting an exhibition?

🎯 Goal

The goal of this project was to create an interactive physical game where users interact with a dollhouse, make choices about sustainability and energy use, and see real-time consequences, while being guided by a friendly energy monster.

👩🏻‍💻 My Contribution

- Led project management and team coordination
- Designed the core interaction concept and user flow
- Created sketches and cardboard prototypes for testing

🔄 Design Process

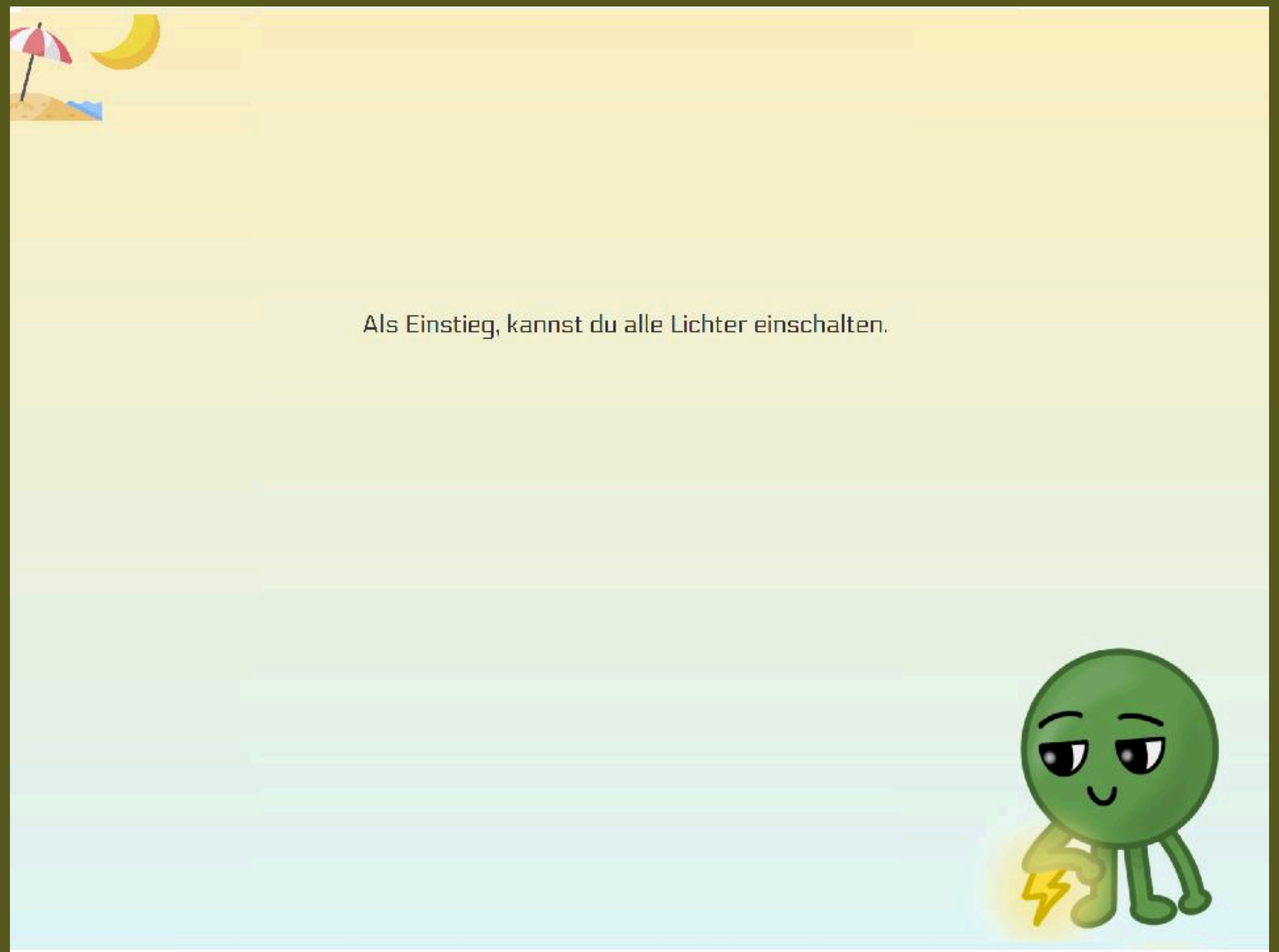
Brainstorming & Concept → Figma prototype → Paper prototype → Implementation → User tesing → Iteration

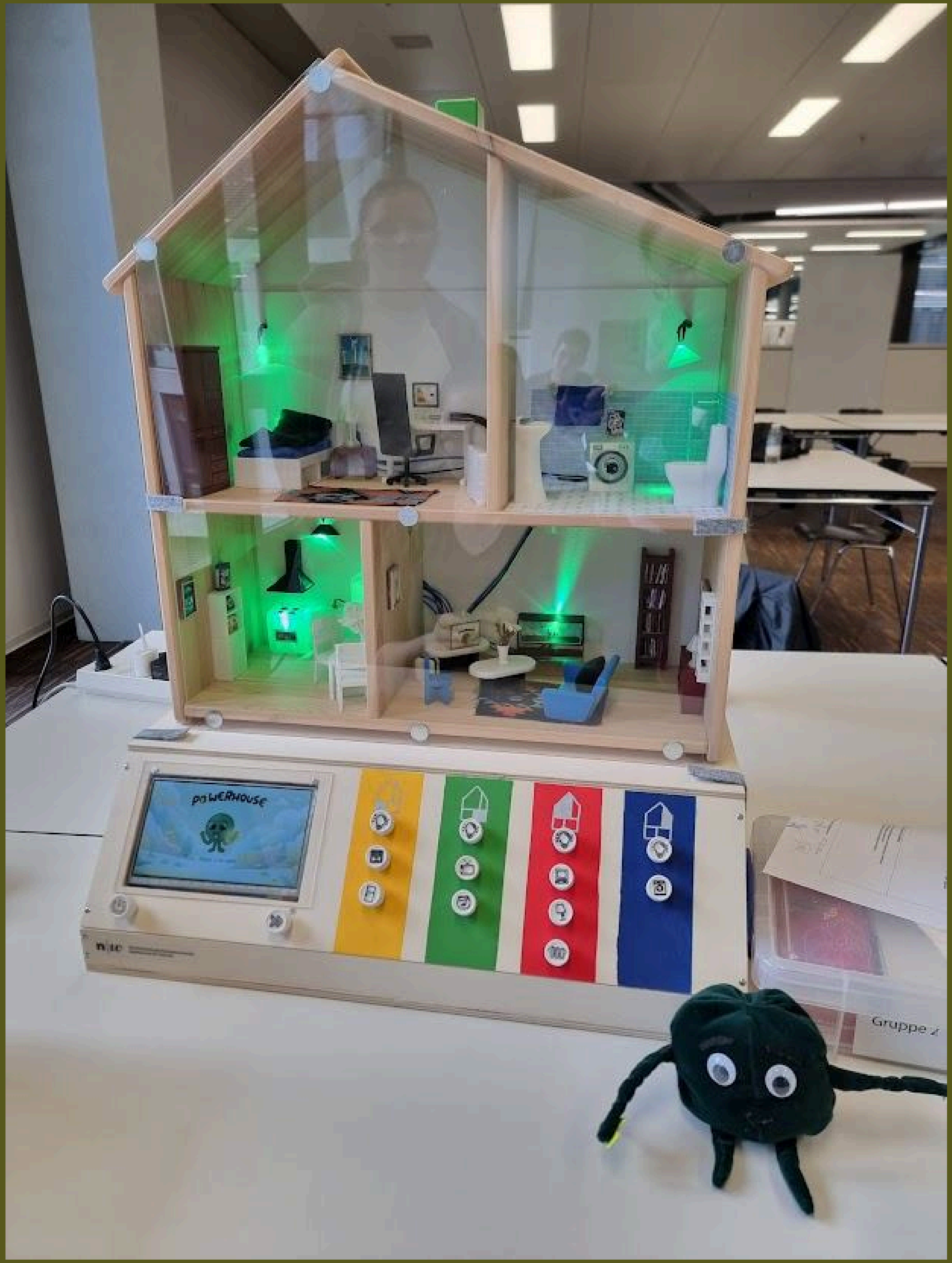
🚀 Key Features

- **Dollhouse:** Physical dollhouse with LEDs that light up when "on"
- **Controls:** Push buttons to toggle appliances + immediate visual feedback
- **Task-Based Play:** Daily scenarios (morning routine, cooking) with energy-saving goals
- **Elektronix Guide:** Energy monster starts exhausted + recharges as you save energy

💡 Reflection

This project taught me how to design for embodied interaction. When you press a button and see a light turn off and the energy meter drop, you feel the impact. Working with a 7-person team also strengthened my project management skills while coordinating hardware, software, and design required clear communication.





🐴 WISP: Connect with your Horse

A smart collar and companion app designed for emotional connection, not just tracking

Module: Advanced Experience Design
Team: 4 people
Duration: 1 semester
Tools: Figma, Miro, ChatGPT (brainstorm features, generating images)

💭 Concept

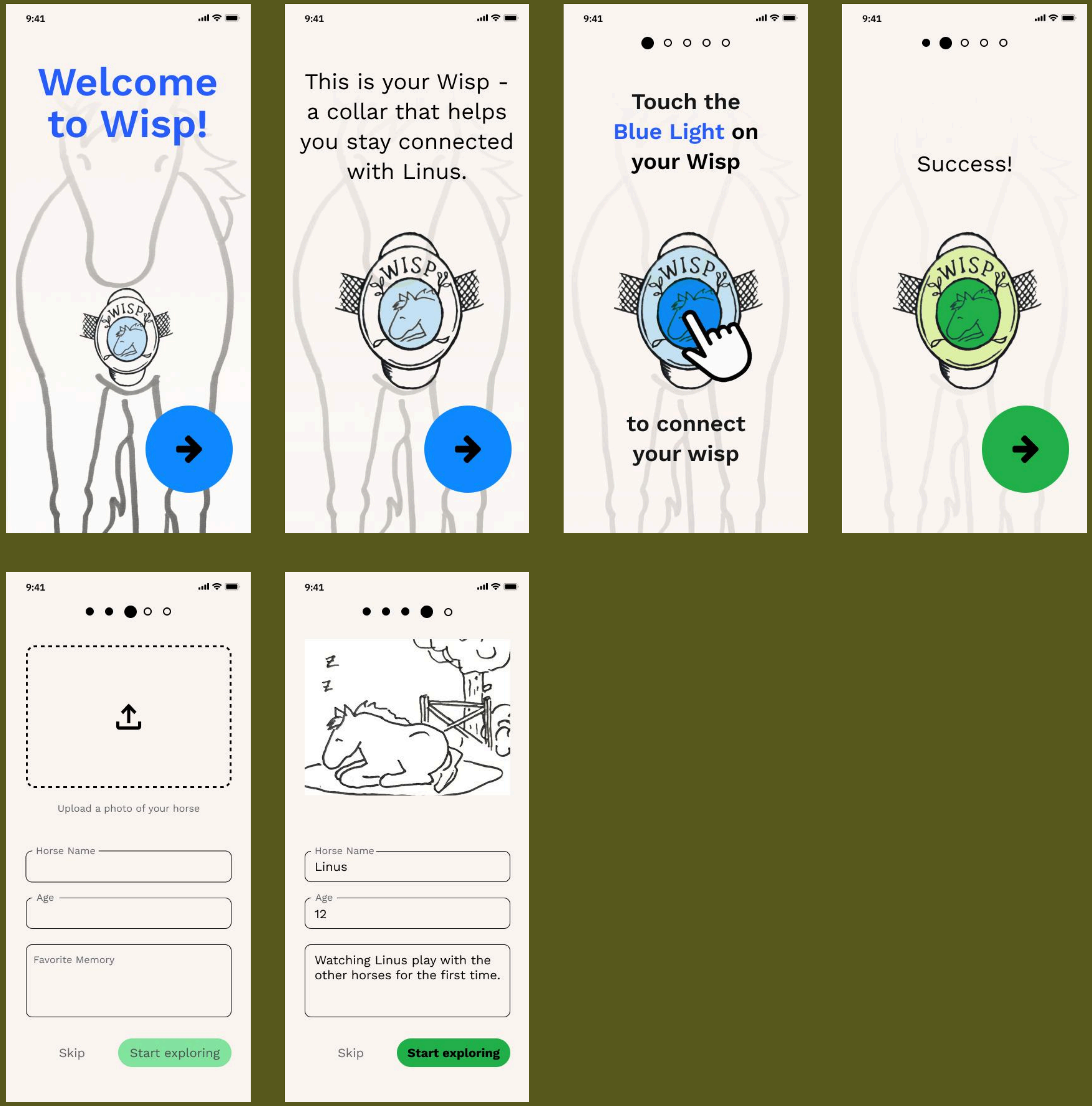
WISP is more than a pet tracker. It's a system designed to maintain and deepen the emotional connection between owners and their horses when they're apart. It pairs a physical smart collar with an app that focuses on presence, memories, and shared moments rather than just data.

🧐 Problem

Horse owners often spend significant time away from their animals due to work, studies, or other commitments. Existing pet tech focuses on health metrics but neglects the emotional bond. Owners miss the small, everyday moments and worry their connection might fade.

👩🏻‍💻 My Contribution

- Translating the conceptual idea into a coherent visual language and interaction design
- Creating the Figma prototype, from onboarding to core features
- Defining the visual design system
- Designing key interaction flows (onboarding, audio messages, reports) to feel intuitive and caring



Onboarding of the app: Connecting the device and setting up the profile.

🔄 Design Process

User interview → Understanding users → Persona → 'How Might We' questions → Brainstorming → Sketches → Storyboards → Figma prototype

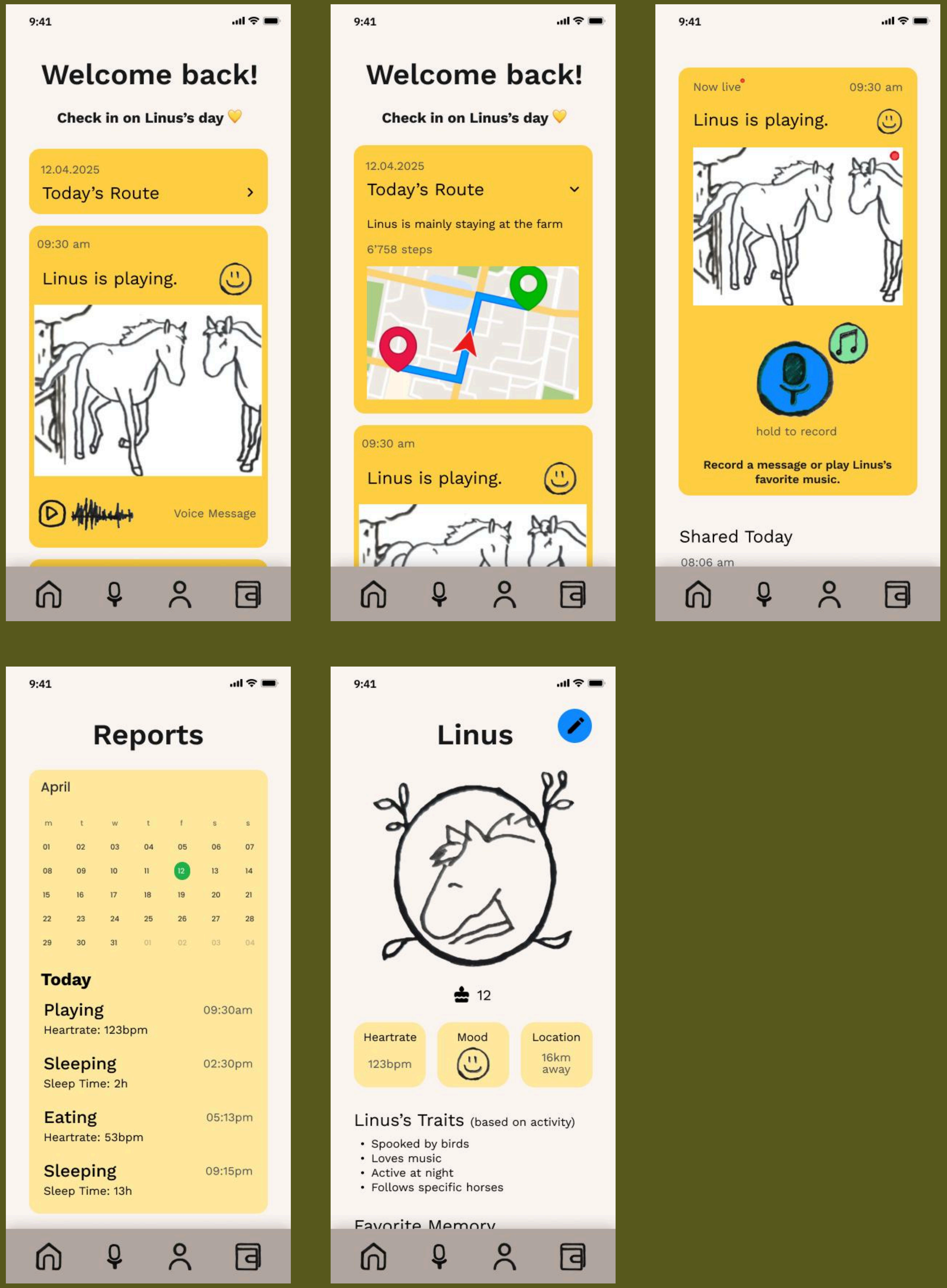
🚀 Key Features

- **Onboarding:** Visual connection feedback (blue → green) + haptic confirmation + add horse details and favorite memory
- **Home:** Today's route + real-time activity notifications + mood tracking + one-tap audio messages
- **Audio:** Live camera feed + hold-to-record + calming sound library
- **Profile:** Health metrics + personality traits + settings
- **Reports:** Calendar view + daily summaries + relive past moments
- **Auto Photos:** Auto-capture when horses meet + cross-device sharing

💡 Reflection

The biggest challenge was balancing meaningful features with simplicity. The solution was to focus every feature on the question: "Does this bring the owner closer to feeling present with their horse?"
If this were to be developed further, I would:

- user-test the prototype with real horse owners to see
- refine the features based on their daily routines



Main features of the app.

🌱 Kitchener – Rewards App

A gamified sustainability app that rewards users for eco-friendly actions

Module: User Interface and Interaction Design
Team: 2 - 4 people (part 1: 4 people; part 2: 2 people)
Duration: 1 semester
Tools: Figma, Milanote, ChatGPT (generating images, brainstorm feature ideas, support with experience vision/persona)

🧐 Problem

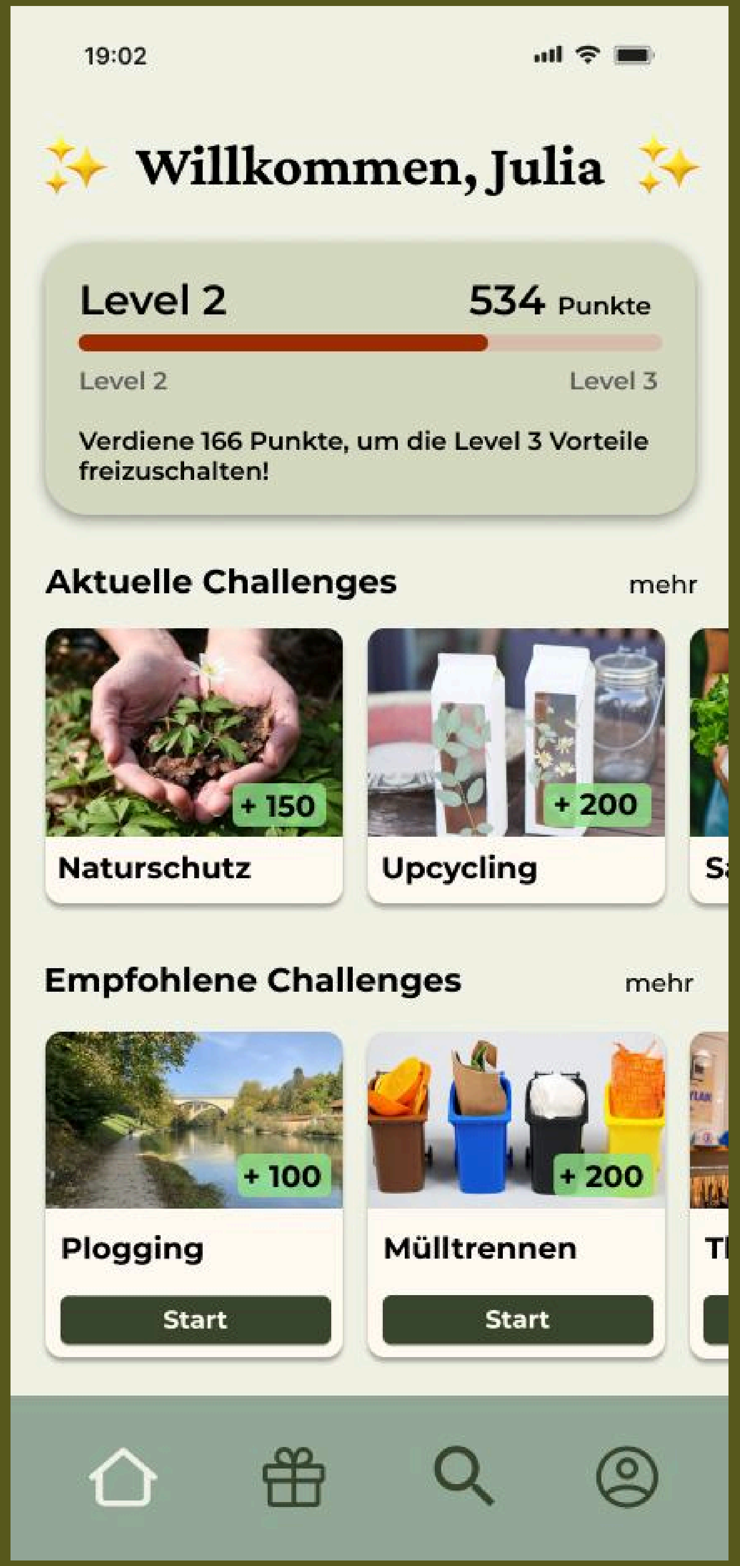
Many people want to live more sustainably, but struggle to stay consistent. Sustainability often feels like sacrifice instead of something enjoyable. There's a lack of motivation, playful elements, and concrete rewards for small, everyday eco-friendly actions.

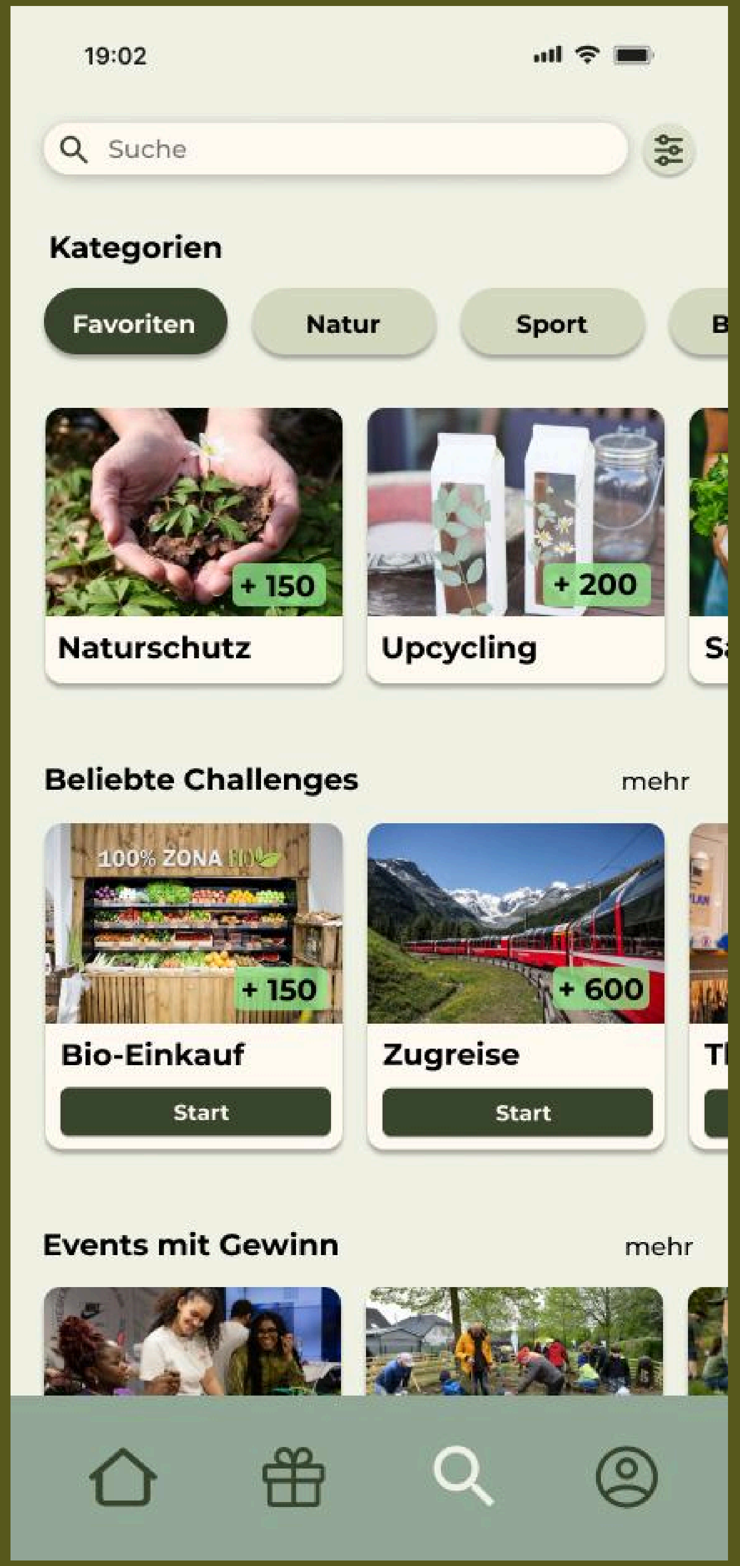
🎯 Goal

Develop an app that makes sustainability feel playful and rewarding. Users stay motivated through a points system, levels, and challenges that integrate eco-friendly habits into their daily lives.

🧑‍💻 My Contribution

- Developed Experience Vision and Persona
- Created lo-fi prototyping with user flow and main screens
- Conducted hallway testing and iterated based on feedback
- Designed moodboard for visual language
- Implemented hi-fi prototyping in Figma
- Ensured consistent visual design aligned with persona





🔄 Design Process

Experience vision → Persona → Lo-Fi Prototyping → User flow → User testing → Moodboard → Hi-Fi Prototyping → Hallway testing

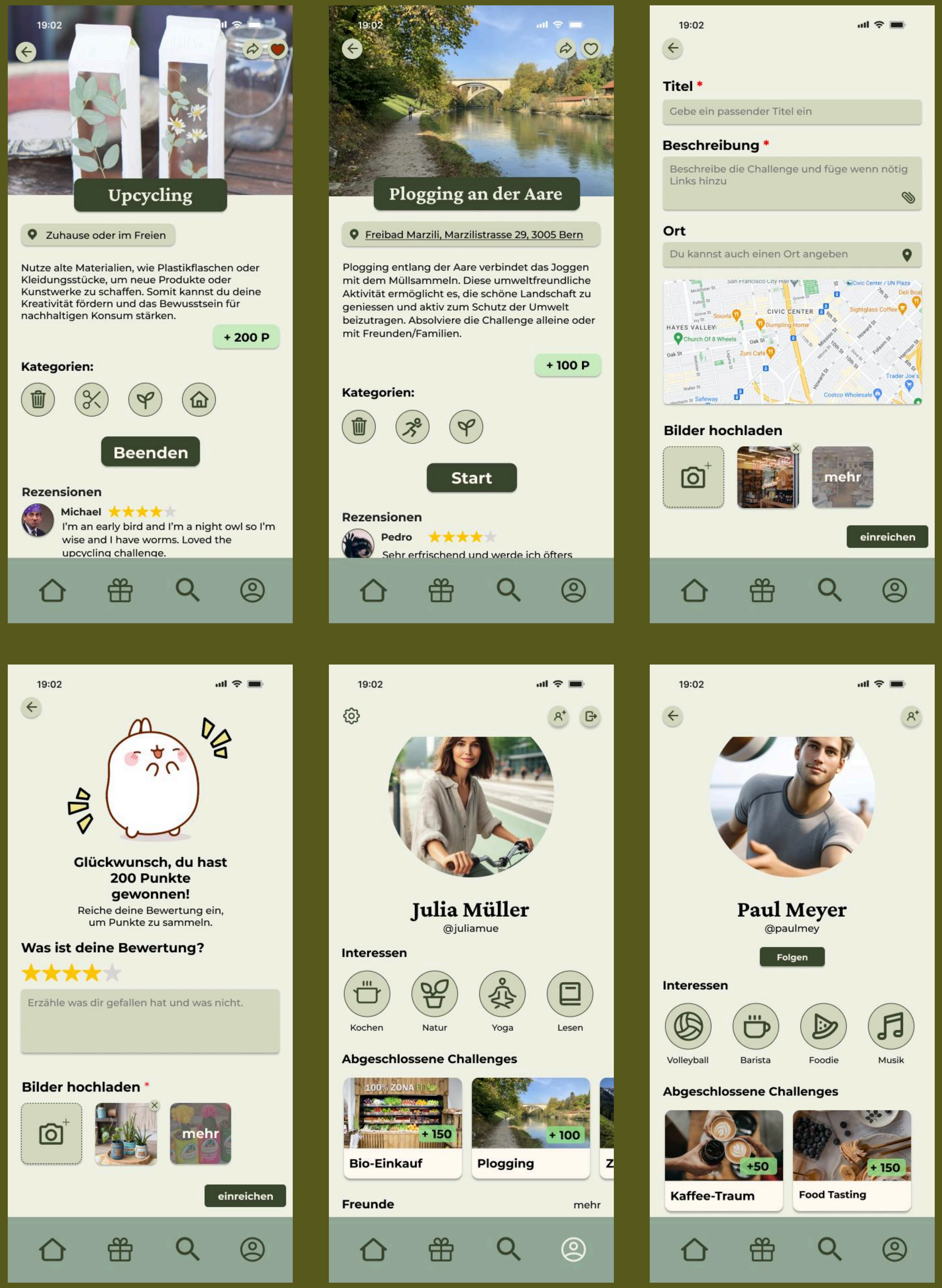
🚀 Key Features

- **Level System:** with new benefits per level
- **Points Tracker:** Progress bar shows points needed for next level
- **Current Challenges:** Active tasks (Nature protection, Upcycling)
- **Recommended Challenges:** Personalized suggestions with point values
- **Challenge Details:** Points, location, review
- **Level Benefits:** Oreview of rewards

💡 Reflection

This project taught me how important a consistent visual language is for conveying a specific emotion. The challenge was making sustainability feel positive and achievable rather than preachy. Gamification was the right approach but hallway testing showed that the connection between action and reward must always be clear, otherwise users lose interest.
If I continued working on this, I would:

- Add more social features (challenge friends)
- Add an onboarding tutorial for the level system



🗃️ Other Projects

🎶 The Evolution of Music – An Interactive Visualization

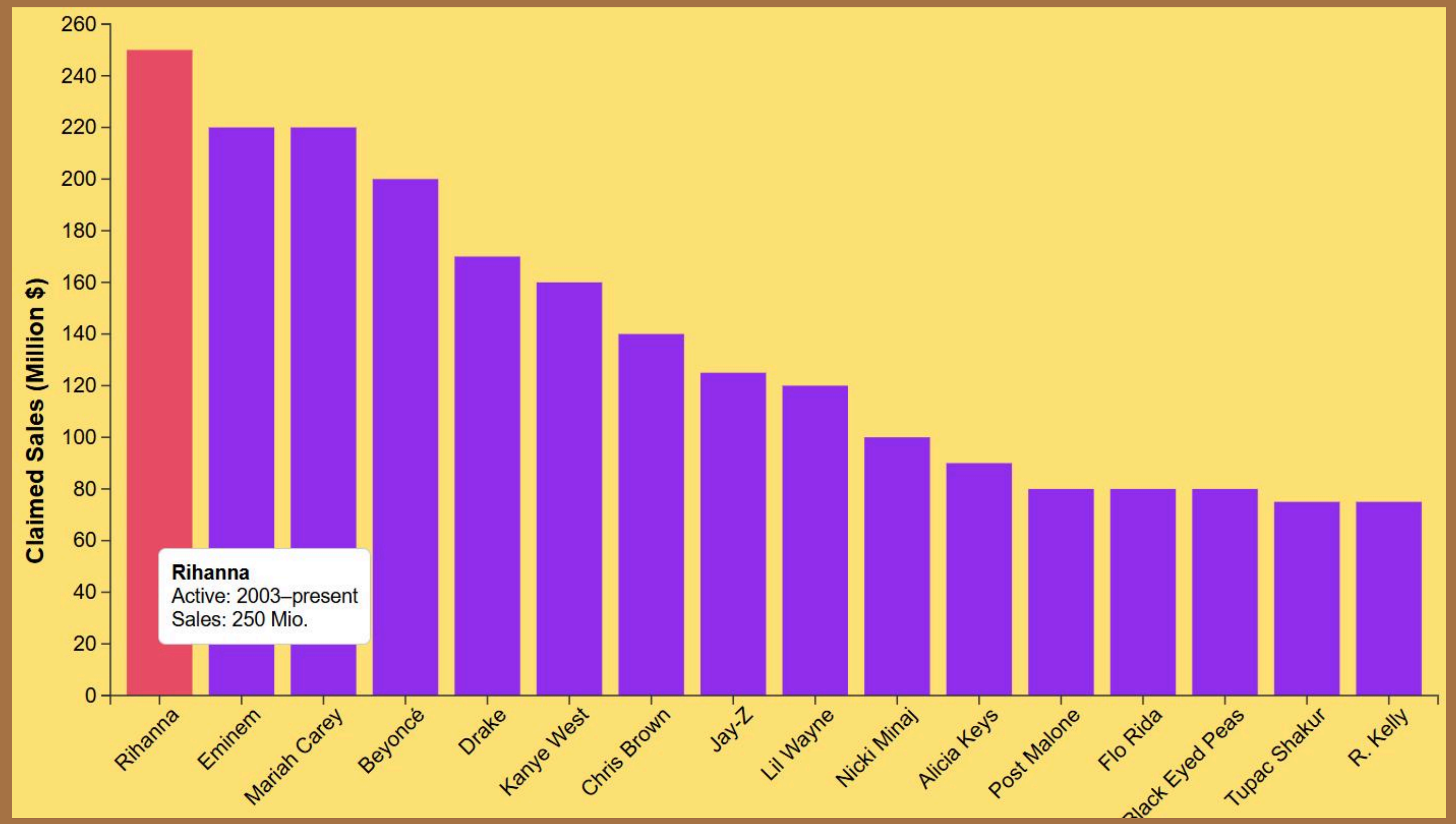
An interactive data visualization that explores the evolution of music through decades, with a focus on Spotify's history. The project presents music data over the years in engaging and explorable formats.

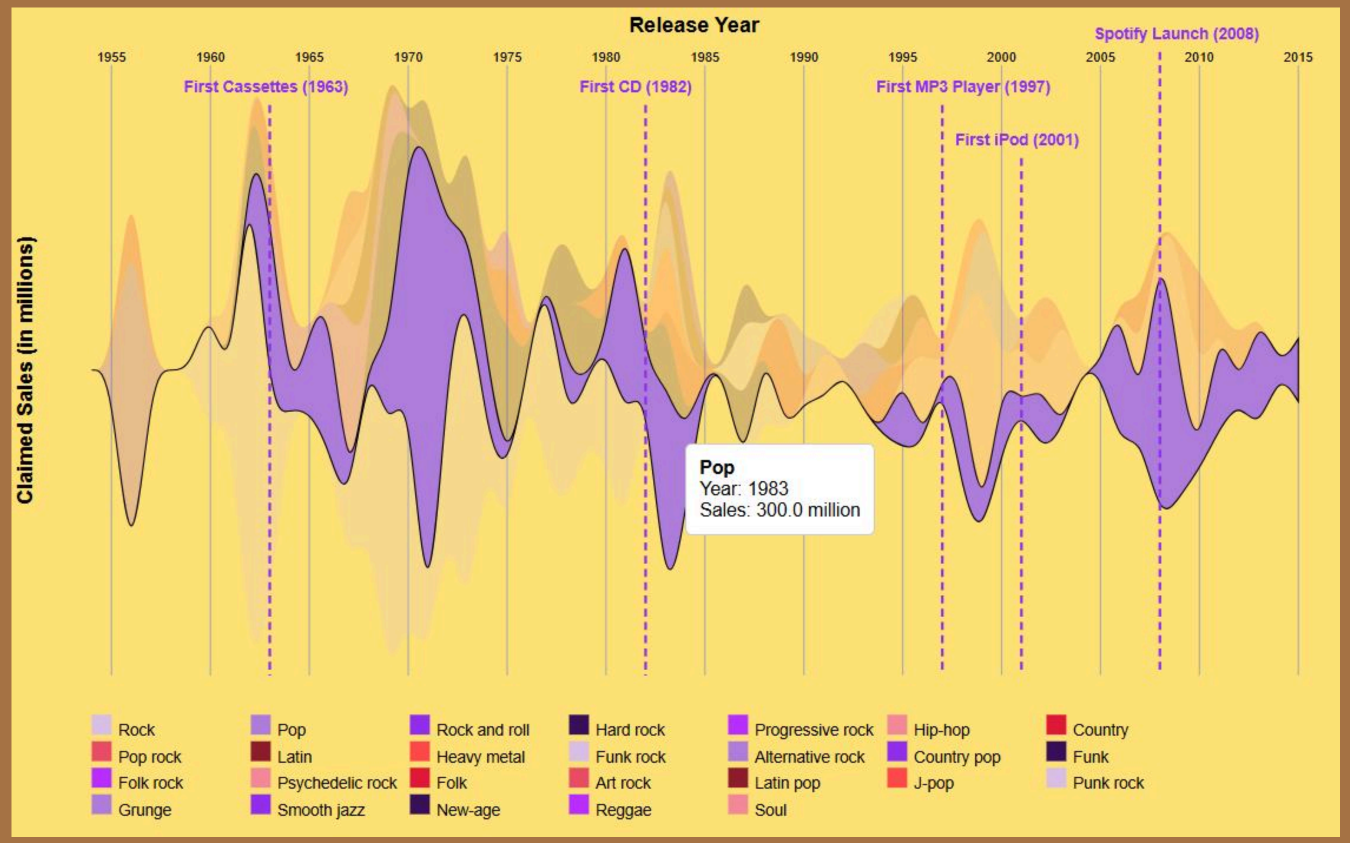
- **Module:** Information Visualization
- **Tools:** Figma, D3.js, HTML, CSS

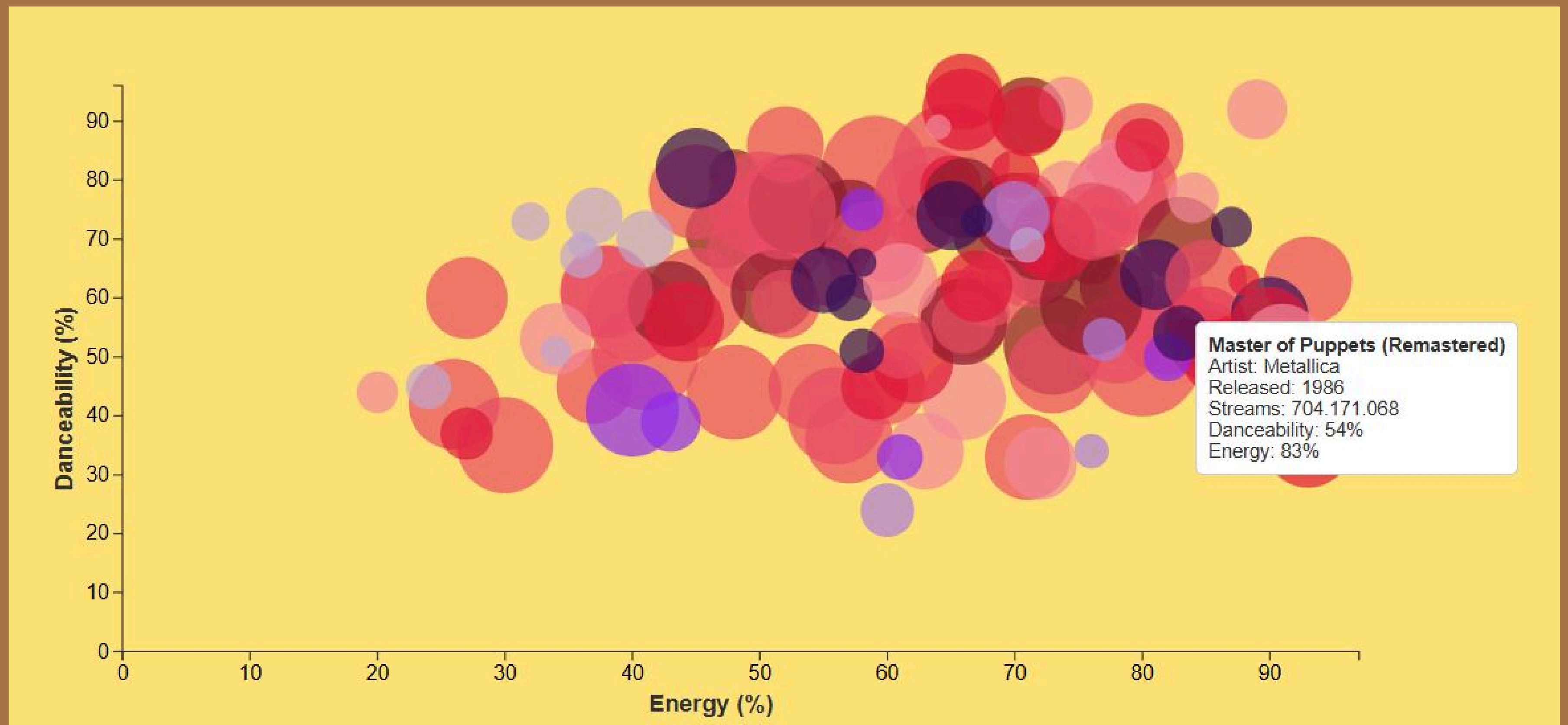
Key features:

- Interactive D3.js visualizations showing music evolution by decade and genre
- Spotify history and facts
- Dynamic charts revealing connections between genres and claimed sales

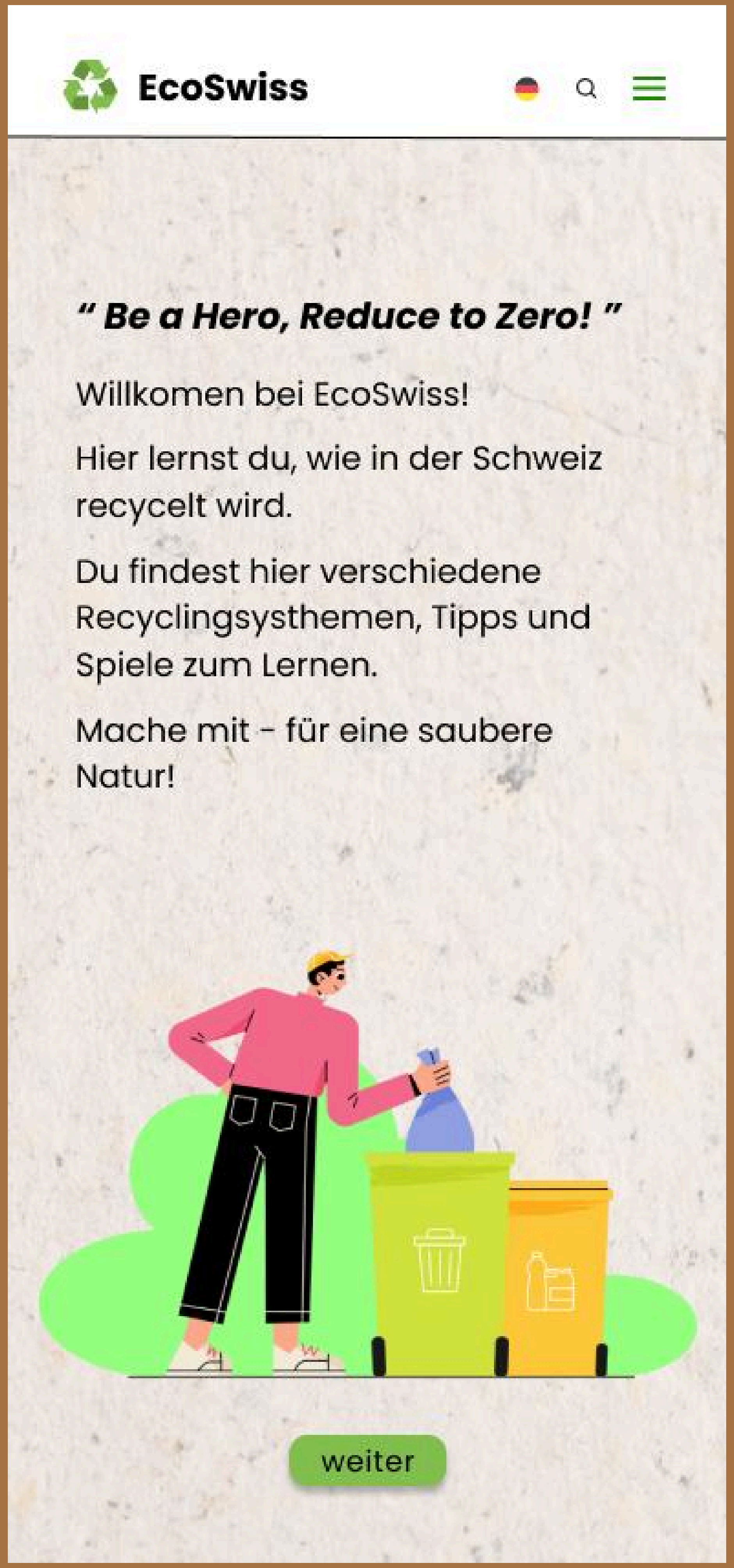
🔗 Live Project







Other Projects



Recycling App

A resourceful app designed to help newcomers in Switzerland understand the country's recycling system with informative guides and games to encourage the recycling lifestyle.

- **Module:** Workshop Design Thinking
- **Tools:** Figma, Miro

Key features:
- Overview of the recycle system in Switzerland
- Mini game to learn about different recycling symbols
- Contact form to reach out to the team

Animal Crossing: New Horizons Fanpage

A Animal Crossing: New Horizons character database that allows users to explore, edit, delete, favorite, and even create new characters in a user-friendly interface.

- **Module:** Web Programming
- **Tools:** JavaScript, HTML, CSS, Grails 6

Key Features:
- Filter by gender and personality
- Favorite villagers list with persistent storage
- Create new villagers with custom attributes

🗂️ Other Projects

👻 Silent Hill 2 Remake – Fanpage

An immersive fan-made website dedicated to the horror game 'Silent Hill 2 Remake'. It presents a gripping overview of the plot, artworks and main characters.

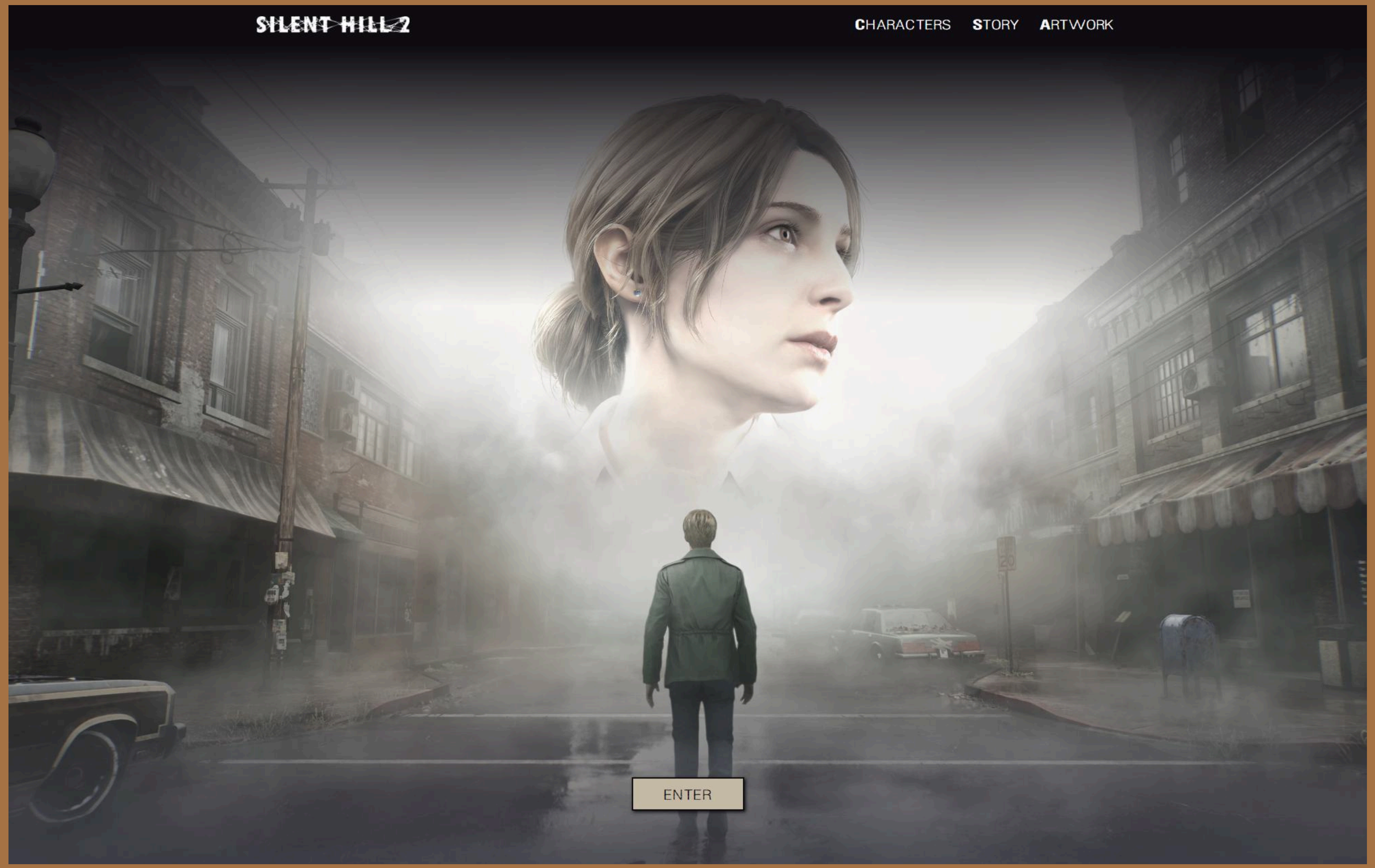
This website is a personal project.

Tools: Figma, HTML, CSS, JavaScript, Gemini (generating code for responsive design)

Key Features:

- Responsive design for mobile/desktop
- Overview and link to the video game
- Gallery of characters and artwork

🔗 Live Project



💌 Love Letter – Webpage

A heartfelt digital space for couples to share memories, love letters, and special moments in a beautifully designed, nostalgic setting.

This website is a personal project.

Tools: Figma, HTML, CSS, JavaScript, Gemini (generating code for responsive design)

Key Features:

- Responsive design for mobile/desktop
- Personal gallery of photos/playlists

🔗 Live Project